1.0 INTRODUCTION

1.1 Project Overview

The Varsity Cyber Clash E-Sports Tournament is a proposed regional competitive gaming event designed to be hosted at Dewan Canselor, Universiti Teknikal Malaysia Melaka (UTeM). The tournament aims to provide a professional, high-energy, and technically sophisticated gaming experience for the university community while serving as a flagship event that showcases the convergence of competitive gaming, broadcast media, and enterprise-grade networking infrastructure.

As the lead technical consulting team for this project, our responsibility is to design, propose, and justify a complete network infrastructure capable of supporting all operational requirements of the tournament — from competitive gameplay and tournament administration to livestream broadcasting and spectator engagement. The network must operate reliably under peak load conditions, deliver low-latency communication for time-sensitive gaming traffic, and provide sufficient bandwidth to support simultaneous media streaming to external audiences.

The event is expected to host 250 concurrent users, distributed across three primary functional groups: competitive gamers (60 participants), administrative and broadcasting personnel (25 staff), and spectators (165 attendees). Each user group presents distinct technical requirements in terms of bandwidth allocation, latency tolerance, security controls, and access privileges. The network design must therefore balance these requirements through intelligent segmentation, prioritization, and resource allocation.

The project was developed in response to the course specification for BAXZ 1313 – Data Communication and Networking, Semester 2, Session 2025/2026, which tasks student teams with preparing a comprehensive event proposal for a university-level E-Sports tournament. The proposal must demonstrate technical competency in network design, cost optimization, and infrastructure planning while remaining practically feasible within a constrained budget.

1.2 Mission and Objectives

1.2.1 Mission Statement

To design and plan a cost-optimized, high-performance enterprise network infrastructure that delivers a seamless competitive gaming experience, supports professional livestream broadcasting, and engages spectators through innovative digital services — all within a controlled, secure, and reliable environment.

1.2.2 Primary Objectives

The Varsity Cyber Clash network design is guided by the following six strategic objectives, each directly addressing a critical operational requirement:

Objective 1: Low-Latency Communication for Competitive Gameplay Gaming traffic must be transmitted with minimal delay, jitter, and packet loss to ensure fair and responsive competitive conditions. This is achieved through VLAN isolation of gaming traffic, Quality of Service (QoS) prioritization, and dedicated high-performance network paths.

Objective 2: High-Bandwidth Availability for Livestream Broadcasting The network must support simultaneous high-definition video streaming from multiple cameras and production workstations to public streaming platforms without degradation of gaming or administrative traffic.

Objective 3: Secure Network Segmentation Through VLAN Implementation Different user groups must be logically isolated to prevent unauthorized access, contain broadcast domains, and enforce security policies. Five dedicated VLANs are deployed to separate Players, Administrators, Servers, Spectators, and Network Management traffic.

Objective 4: Reliable Wireless Connectivity for Spectators and Staff The wireless infrastructure must accommodate 165 spectators and event staff with stable connectivity, sufficient per-user bandwidth, and secure authentication Six managed wireless access points will be deployed to provide overlapping coverage throughout the venue.

Objective 5: Cost Minimization Through Strategic Procurement The total project cost must be minimized without compromising performance, reliability, or scalability. This is achieved through a hybrid procurement model that combines the rental of end-user devices (gaming PCs, workstations, servers) with the purchase of long-term network infrastructure (switches, router, access points).

Objective 6: Scalability and Future Expansion The network architecture must be designed using a hierarchical model that allows for future expansion in user count, bandwidth demand, or service offerings without requiring fundamental redesign.

1.3 Project Scope

1.3.1 In-Scope Activities

The scope of this project encompasses the following activities and deliverables:

No.	Activity	Description
1	Venue Selection and Justification	Identification and justification of Dewan Canselor, UTeM, as the tournament venue based on spatial, power, and cooling criteria.
2	Network Design	Development of a Three-Tier Hierarchical Star Topology with VLAN segmentation and Inter-VLAN routing.
3	IP Addressing Scheme	Design of a VLSM-based private IP addressing plan for all network segments.
4	Cost Analysis	Comprehensive cost estimation for hardware, infrastructure, services, and manpower.
5	Procurement Planning	Identification of rental and purchase sources for all required equipment.
6	Technical Portfolio	Documentation of hands-on competency in cable construction, IP configuration, and wireless deployment.
7	Additional Service Design	Proposal of innovative services to enhance spectator engagement and project complexity, as detailed in Section 3.6.

1.3.2 Out-of-Scope Activities

The following activities are explicitly outside the scope of this proposal:

- Procurement of gaming licenses, game software, or game-specific subscriptions.
- Tournament prize money, marketing, and event promotion.
- Catering, accommodation, and participant travel arrangements.

- Post-event equipment disposal, recycling, or long-term storage.
- Detailed cybersecurity penetration testing or compliance auditing.

1.3.3 Project Boundaries

The project focuses on the network infrastructure and supporting systems required to operate the tournament. It does not include the organization of the tournament itself (scheduling, rules, refereeing), which is managed by a separate organizing committee.

1.4 Report Structure

This report is organized into nine major sections, each addressing a specific aspect of the proposed tournament infrastructure. Collectively, these sections provide a comprehensive analysis of venue planning, network architecture, IP addressing, cost optimization, technical implementation, project complexity, and supporting documentation. The structure is designed to align with the project requirements while ensuring logical progression from project planning and design to costing, evaluation, and final recommendations.

2.0 LOGISTIC DETAILS

2.1 Venue Selection and Justification

2.1.1 Selected Venue

Dewan Canselor, Universiti Teknikal Malaysia Melaka (UTeM), has been selected as the official venue for the Varsity Cyber Clash E-Sports Tournament. This venue was identified after a systematic evaluation of available halls and open spaces on the UTeM main campus based on three primary criteria: **spatial capacity, existing power infrastructure, and**

cooling capabilities — exactly the three justification factors required by the project specification.

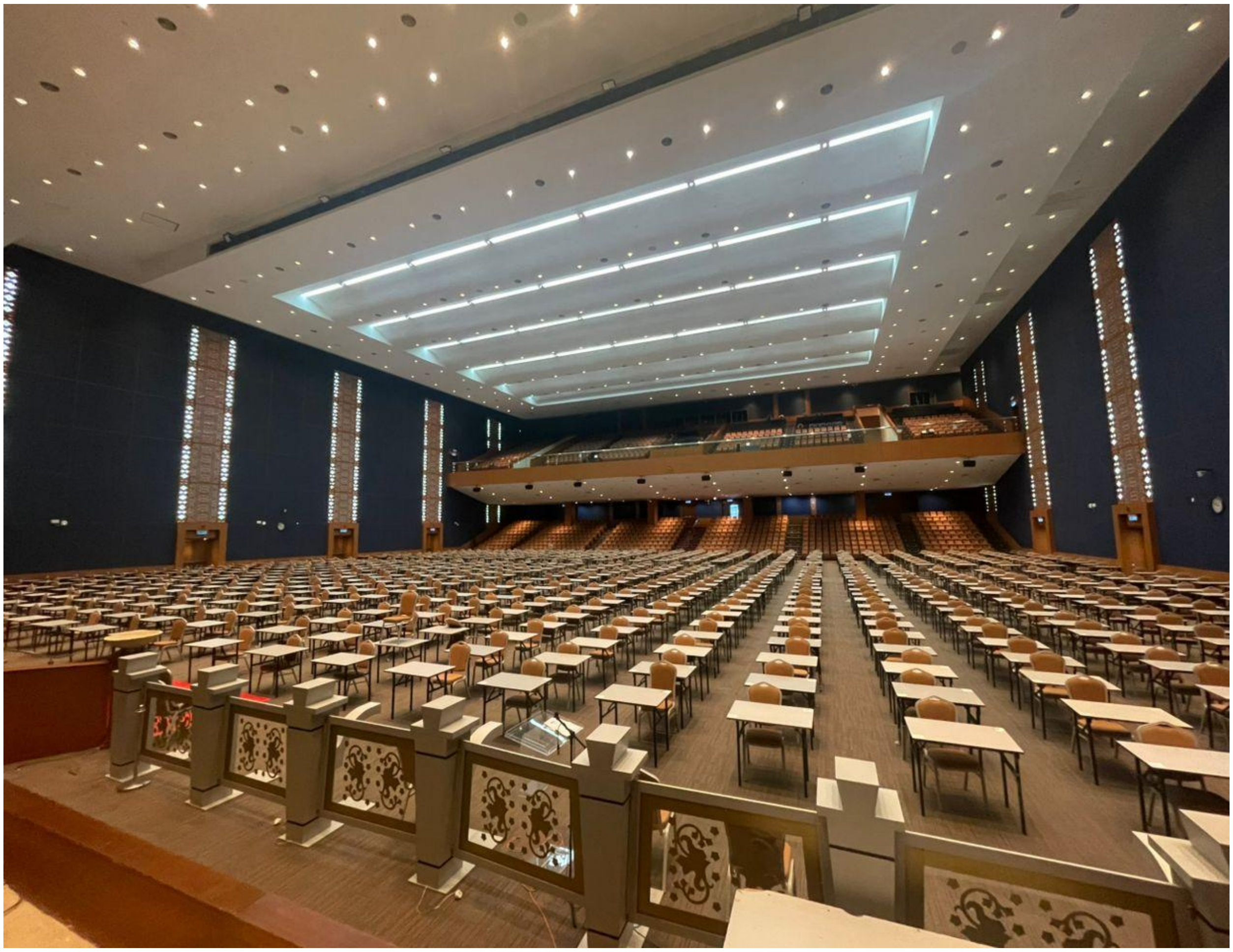
2.1.2 Strategic Location

Dewan Canselor is located at the **central core of the UTeM main campus in Durian Tunggal, Melaka**. Its strategic position provides several operational advantages:

- **Centralized accessibility** for students, staff, and external visitors from all faculties.
- **Proximity to public transport routes** and main campus parking facilities, supporting high-volume spectator arrival and departure.
- **Integration with existing campus services** including security patrols, medical first-aid stations, and cleaning crews, all of which can be extended to cover the event.
- **High visibility** within the campus, supporting event promotion and spectator turnout.

2.1.1 Figure— photo showing the strategic location of Dewan Canselor on the UTeM campus

2.1.3 Spatial Capacity Justification



Dewan Canselor provides more than **650 square meters of usable floor space** on its ground floor and is officially rated to accommodate approximately **800 to 1,000 occupants** in a theater or banquet configuration.

The Varsity Cyber Clash tournament requires space for the following functional zones:

- 60 competitive gaming stations (players)
- 25 administrative and broadcasting workstations
- 165 spectator seats
- 6 enterprise servers and supporting IT infrastructure
- Centralized network equipment rack

- Walkways, emergency access, and circulation space

Since the venue's maximum capacity (800–1,000) substantially exceeds the tournament's expected attendance (250 users), the venue comfortably satisfies all spatial requirements while providing **substantial reserve capacity** for:

- Equipment staging and storage areas
- Emergency evacuation buffers
- Future participant expansion
- Comfortable crowd movement and reduced congestion

This capacity margin directly supports the project's **safety, comfort, and operational flexibility** objectives.

2.1.4 Existing Power Infrastructure Justification

Dewan Canselor is equipped with a **three-phase industrial electrical distribution system** designed to support high-consumption events such as convocations, exhibitions, and concerts. The venue provides:

- Multiple 13A and 15A power outlets distributed along wall perimeters.
- Dedicated three-phase power input suitable for high-load equipment.
- Existing main distribution board (MDB) with sufficient breaker capacity for temporary sub-distribution.

The estimated total electrical load of the tournament (approximately 80 kW) represents a manageable draw on the venue's distribution system, as detailed in Section 2.3. The use of the venue's existing electrical infrastructure eliminates the need for generator rental or major electrical upgrades, directly contributing to the project's **cost-minimization objective**.

2.1.5 Existing Cooling Capabilities Justification

The venue is equipped with a **centralized air-conditioning system** capable of maintaining indoor temperatures between 20°C and 24°C under normal occupancy loads. This temperature range is critical for the operational stability of high-performance gaming PCs, enterprise servers, and networking equipment, all of which are sensitive to thermal throttling at elevated temperatures.

The existing cooling capacity has been verified (through venue management consultation) to support an event of this scale. As a supplementary measure, the project budget includes high-velocity industrial fans (within the electrical setup allocation) to provide localized airflow within the gaming and server zones. This hybrid cooling approach ensures continuous thermal management without requiring the rental of portable industrial air-conditioning units, which would significantly increase the project cost.

2.1.6 Decision Summary

Criterion	Venue Capability	Tournament Requirement	Margin Status
Floor Space	>650 m^2	~450 m^2 (Planned Allocation)	✓ +200 m^2 reserve
Occupancy Capacity	800–1,000 persons	250 persons	✓ +550 reserve
Power Distribution	Three-phase industrial	80 kW total	✓ Adequate
Cooling System	Centralized AC (20–24°C)	Required for IT equipment	✓ Adequate
Accessibility	Central campus	High public access	✓ Optimal

Conclusion: Dewan Canselor, UTeM, satisfies all primary venue requirements with substantial operational margins, and is therefore the optimal and most cost-effective choice for the tournament.

2.2 Space Allocation and Capacity Planning

2.2.1 Zoning Philosophy

The venue floor has been divided into **five functional zones** based on the operational, security, and technical requirements of each user group. The zoning strategy follows three design principles:

1. **Logical separation** of high-priority gaming traffic from public spectator traffic.
2. **Centralized placement** of network equipment to minimize cabling distance and reduce infrastructure cost.

3. **Compliance with safety regulations** regarding emergency exits, fire lanes, and crowd movement.

2.2.2 Zone Definitions

Zone 1: Players Zone (Gaming Arena) This zone accommodates the 60 competitive gamers arranged in a stadium-style layout facing the main stage. Each gaming station consists of a custom gaming PC, monitor, peripherals, and a dedicated seat. Approximately **90 square meters** are allocated to provide adequate spacing between stations, comfortable seating, and clear sightlines for referees and broadcast cameras.

Zone 2: Administration and Tournament Operations Zone This zone supports all back-of-house functions including participant registration, referee coordination, technical support, and event management. It contains 25 administrative workstations, a registration desk, and a referee station. Approximately **40 square meters** are allocated to support workflow efficiency and confidential operations.

Zone 3: Broadcast and Technical Support Area This dedicated area houses the streaming equipment, production workstations, network monitoring systems, and the centralized server rack. It is positioned adjacent to the main stage to facilitate camera-to-server signal flow. Approximately **30 square meters** are allocated.

Zone 4: Spectator Area A theater-style seating arrangement accommodates the 165 spectators with clear visibility of the main stage and gaming screens. Approximately **165 square meters** are allocated to ensure comfortable seating, clear aisles, and unobstructed sightlines.

Zone 5: Walkways, Safety Corridors, and Equipment Staging Approximately **80 square meters** are reserved for circulation, emergency access routes compliant with fire safety codes, equipment staging, and future adjustments.

2.2.3 Space Allocation Table

No.	Zone	Estimated Space (m^2)	Percentage of Recommended Allocation
1	Players Zone	90	20.0%
2	Administration & Officials Zone	40	8.9%

No.	Zone	Estimated Space (m^2)	Percentage of Recommended Allocation
3	Broadcast & Technical Area	30	6.7%
4	Spectator Area	165	36.7%
5	Walkways & Safety Corridors	80	17.8%
Total Operational Requirement		**405**	**90.0%**
Planned Allocation (including contingency)		**450**	**100%**
Venue Available Capacity		**>650**	**144%+**

2.2.4 Layout Strategy

[IMAGE: Figure 2.2.1 — Floor plan / layout diagram showing the five zones and their positions within Dewan Canselor]

The proposed layout positions the **Players Zone adjacent to the main stage**, allowing spectators to directly view both the players and the main display. The **Broadcast and Technical Area** is placed at the rear of the main stage to enable short cable runs to cameras and to the central network equipment rack. The **Administration Zone** is positioned near the venue entrance to facilitate participant check-in flow. The **Spectator Area** occupies the rear two-thirds of the venue with tiered-style seating. This arrangement minimizes cable length (reducing infrastructure cost), optimizes broadcast angles, and ensures emergency egress is not obstructed.2.3 Power Infrastructure

2.3.1 Total Electrical Load Requirement

The tournament requires a stable electrical capacity of approximately **80 kW** to support gaming systems, servers, networking equipment, broadcasting infrastructure, and supporting services. A reserve margin is included to prevent circuit overload and maintain operational stability throughout the event.

2.3.2 Load Calculation Table

The total electrical load is calculated by summing the peak power consumption of every powered device in the tournament infrastructure, as detailed in Table 2.3.1.

No.	Equipment Category	Quantity	Unit Power (W)	Total Power (W)	Total Power (kW)
1	Custom Gaming PCs (i5-14400F, RTX 4060)	60	800	48,000	48.0
2	24-inch Full HD Monitors	85	40	3,400	3.4
3	Dell PowerEdge T620 Servers	6	1,500	9,000	9.0
4	Network Equipment (Router, Switches, APs)	10	200	2,000	2.0
5	Broadcasting Equipment (Cameras, Mixers, Encoders)	8	800	6,400	6.4
6	Streaming Workstations (4 units)	4	650	2,600	2.6
7	Subtotal (Active load)	—	—	71,400	71.4
8	Safety Margin(12.04)%	—	—	8,600	8.6
Total Estimated Load				**80,000 W**	**80.0 kW**

Calculation Methodology:

- Gaming PCs are rated at 800W peak (CPU + GPU + peripherals under full competitive load).
- Server power is rated at 1.5 kW under typical enterprise load (per Dell PowerEdge T620 specifications).
- The safety margin (8.6 kW) accounts for peak surge, additional peripherals, and unforeseen load.

2.3.3 Power Distribution Strategy

To balance the 80 kW load safely across the venue's existing distribution system, the following distribution strategy is implemented:

- **Three-phase load balancing**: The total load is distributed across the three phases of the venue's main distribution board to prevent phase imbalance.
- **Dedicated sub-distribution panel**: A temporary 3-phase sub-panel is installed near the central network rack to feed the gaming, server, and broadcast zones.
- **Power extension and cable management**: 25 heavy-duty power extension sockets (rated 13A) are deployed to reach all device locations, with cable trunking to prevent tripping hazards.
- **UPS backup for critical systems**: Two 1500VA UPS units are installed to protect the core router, core switch, servers, and broadcasting encoders from momentary power loss or voltage fluctuation.

2.3.4 Power Infrastructure Cost

The detailed cost of the electrical setup is documented in **Section 4.5.** The total electrical setup cost is **RM 3,675.00**, including power extensions, UPS units, and circuit protection equipment.

2.4 Environmental Cooling

2.4.1 Cooling Strategy

The tournament requires a controlled thermal environment to ensure the operational stability of high-performance gaming hardware, enterprise servers, and networking equipment. Elevated temperatures cause **thermal throttling**, which reduces CPU and GPU clock speeds and directly degrades gaming performance. Servers operating above their rated temperature may experience premature hardware failure or unexpected shutdown.

The cooling strategy employs a **two-layer approach**:

1. **Primary cooling**: The venue's centralized air-conditioning system maintains the ambient temperature between **20°C and 24°C**.
2. **Supplementary cooling**: Strategically positioned **high-velocity industrial fans** within the gaming and server zones provide continuous airflow and prevent hot-spot formation.

2.4.2 Cooling Capacity Requirement

The total heat load to be dissipated is calculated from the electrical load:

Heat Source	Electrical Equivalent (kW)	Thermal Output (kW)
Gaming PCs	48.0	~48.0
Servers	9.0	~9.0
Monitors	3.4	~3.4
Subtotal (Active Thermal load)	—	~96.4
Safety Margin(10%)	—	~9.6
Network Equipment	2.0	~2.0
Broadcasting Equipment	6.4	~6.4
Occupancy (250 persons × 100W)	—	25.0
Total Design Heat Load		**~106 kW**

A standard centralized air-conditioning system rated at

3.5 kW per ton (1 TR = 3.517 kW) requires approximately

30 tons of cooling capacity to dissipate the 106 kW design

load, which is well within the venue's 50 TR HVAC capacity

(≈175 kW cooling output, as per Appendix I).

2.4.3 Cooling Cost Efficiency

The hybrid approach (centralized AC + supplementary fans) is significantly more cost-effective than renting portable industrial cooling units, which would cost an additional RM 3,000–5,000 for the two-day event. The existing venue infrastructure is therefore leveraged to its maximum extent, supporting the project's cost-minimization objective.

2.5 Safety and Emergency Considerations

2.5.1 Emergency Egress

Dewan Canselor is equipped with **multiple emergency exits** distributed around the perimeter of the hall. The proposed venue layout explicitly reserves 80 m^2 for walkways and safety corridors to ensure unobstructed egress in compliance with fire safety regulations.

2.5.2 Fire Safety

The venue is equipped with fire extinguishers, fire alarm systems, and emergency lighting. The proposed equipment layout maintains a minimum clearance of **1.5 meters** from all emergency exits and fire-fighting equipment.

2.5.3 Electrical Safety

All temporary electrical installations are protected by **circuit breakers and residual current devices (RCDs)** as part of the electrical setup budget. Cable trunking is used throughout the venue to protect cables from physical damage and to prevent tripping hazards.

2.5.4 Medical and First Aid

A first-aid station is established within the Administration Zone, staffed by trained personnel. The venue's proximity to the UTeM Health Centre provides additional medical support if required.

2.5.5 Crowd Management

The Spectator Zone is designed with clear aisles and designated entry/exit points to facilitate orderly crowd movement before, during, and after the event.

2.6 Section Summary

The logistic planning for the Varsity Cyber Clash E-Sports Tournament is based on a **systematic venue justification methodology** that addresses each of the three criteria specified in the project specification: spatial dimensions, power infrastructure, and cooling capabilities. Dewan Canselor, UTeM, has been selected as the optimal venue because it satisfies all requirements with substantial operational margins while requiring no major infrastructure investment.

The proposed space allocation strategy divides the venue into five functional zones, with a total operational footprint of **405 m^2** (recommended allocation **450 m^2**) — well within the venue's available **650+ m^2** capacity. This approach optimizes the use of existing infrastructure, supports the cost-minimization objective, and ensures compliance with safety regulations.

The **80 kW total electrical load** is calculated transparently and distributed safely using existing venue power infrastructure supplemented by temporary distribution equipment. The **hybrid cooling strategy** (centralized AC + industrial fans) maintains optimal operating temperatures without the expense of portable cooling units.

Collectively, these logistic decisions form the operational foundation upon which the network architecture (Section 3) and costing strategy (Section 4) are built. Every design decision in subsequent sections is traceable to a logistic constraint or opportunity documented in this section.

3.0 NETWORK ARCHITECTURE AND DESIGN

3.1 Network Topology

3.1.1 Topology Selection and Justification

The Varsity Cyber Clash network infrastructure is built upon a **Three-Tier Hierarchical Star Topology**, which is the industry-standard model for enterprise networks requiring scalability, fault isolation, and simplified management. The three tiers are:

1. **Core / Distribution Layer** — Centralized routing and high-speed backbone.
2. **Access Layer** — End-device connectivity and port density.
3. **Edge Layer** — External network gateway and security boundary.

This topology was selected over alternatives (flat star, mesh, bus) for the following justified reasons:

Alternative	Why Rejected
Flat Star Topology	No scalability, single point of failure, no traffic segmentation
Mesh Topology	Over-engineered for 250 users, excessive cabling cost, complex management
Bus Topology	Obsolete, no fault isolation, poor performance under load
Ring Topology	Single point of failure disrupts entire network

The hierarchical star offers **modular expansion**, **broadcast domain containment** through VLANs, and **clear troubleshooting boundaries** between layers.

3.1.2 Device Placement by Tier

Tier	Device	Quantity	Function
Edge (Router)	TP-Link ER7206 Gigabit VPN Router	1	Internet gateway, NAT, firewall, VPN
Core / Distribution (L3 Switch)	FS S5800-48F4SR Layer 3 Switch	1	Inter-VLAN routing, core switching, 10Gb backbone
Access (L2 Switches)	TP-Link TL-SG3452 JetStream 48-Port Managed	2	End-device connectivity, VLAN tagging, QoS
Wireless	TP-Link TL-WA901ND Wireless Access Points	6	Spectator and staff wireless connectivity
Servers	Dell PowerEdge T620	6	Game, web, FTP, DB, streaming, auth services
End Devices	Custom Gaming PCs + Dell Vostro 3681	60 + 25	User workstations

3.1.3 Topology Diagram

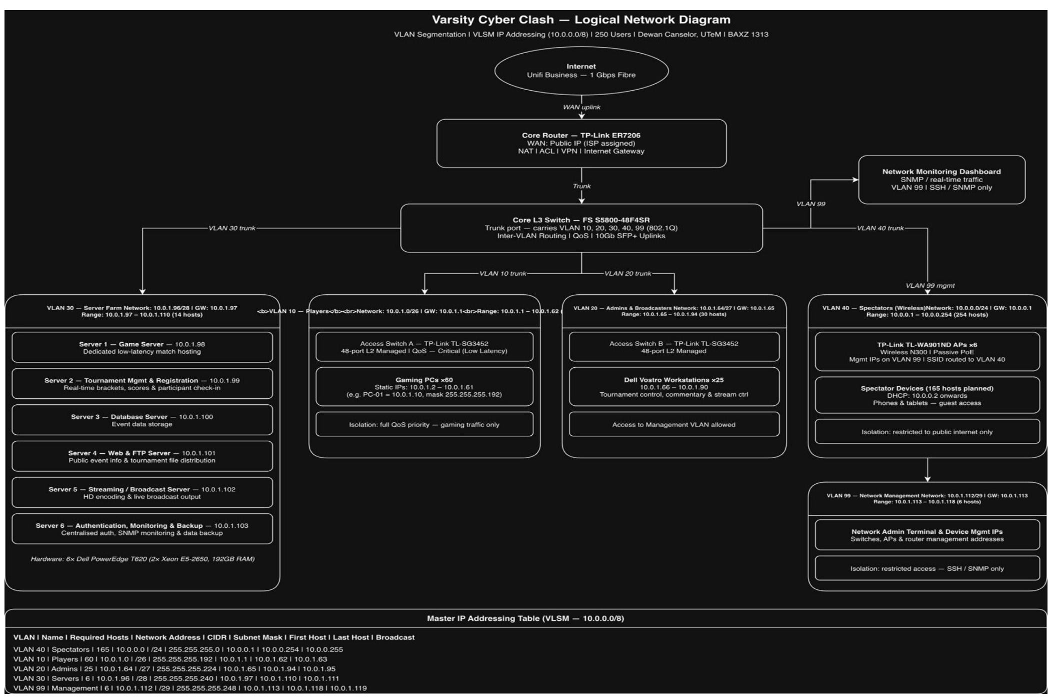


Figure 3.1.1 — Three-Tier Hierarchical Star Topology diagram showing the Internet cloud → ER7206 Router → S5800-48F4SR Core L3 Switch → 2× TL-SG3452 Access Switches → All end devices, servers, and 6 wireless APs]

The diagram shows the **logical placement** of all devices with their connections:

- **WAN uplink** from the venue's fiber internet (1 Gbps Unifi Business) into the ER7206 router.
- **Single 10Gb SFP+ uplink** from the router to the core L3 switch.
- **Two 10Gb SFP+ uplinks** from the core L3 switch to the two access L2 switches (one per switch, providing high-speed uplink connectivity between the core and access layers).
- **Gigabit copper connections** from access switches to all wired end devices (gaming PCs, admin workstations, servers, APs).
- **Logical VLAN segmentation** indicated by color coding (VLAN 10/20/30/40/99).

3.2 VLAN Segmentation Strategy

3.2.1 Segmentation Rationale

Network segmentation is implemented using **Virtual Local Area Networks (VLANs)** to achieve four critical objectives:

1. **Security isolation** — Prevent unauthorized access between user groups (e.g., spectators cannot access gaming PCs).
2. **Broadcast domain containment** — Limit broadcast traffic to relevant user groups, reducing unnecessary overhead.
3. **Performance prioritization** — Apply Quality of Service (QoS) policies per VLAN based on traffic criticality.
4. **Simplified management** — Apply policies and troubleshooting at the VLAN level rather than per-device.

3.2.2 VLAN Allocation Table

VLAN ID	Purpose	User Group	Traffic Priority	Isolation Strategy
VLAN 10	Players	60 Gaming PCs	Critical (Low Latency)	Full isolation with QoS prioritization

VLAN ID	Purpose	User Group	Traffic Priority	Isolation Strategy
VLAN 20	Administrators & Broadcasters	25 Workstations	High	Access limited to authorized services and resources
VLAN 30	Server Farm	6 Servers	High	Restricted to server-to-server and server-to-admin communications
VLAN 40	Spectators	165 Wireless Devices	Standard	Restricted to public internet access only, isolated from internal services
VLAN 50	Production/Cameras	4 IP Cameras+ NVR	High	Isolated broadcast VLAN
VLAN 99	Network Management	Network Devices	Administrative	Restricted access for authorized network administrators only

3.2.3 Inter-VLAN Routing

Inter-VLAN communication is performed exclusively by the **FS S5800-48F4SR Layer 3 Switch** through SVIs (Switched Virtual Interfaces). The following controlled communication paths are permitted:

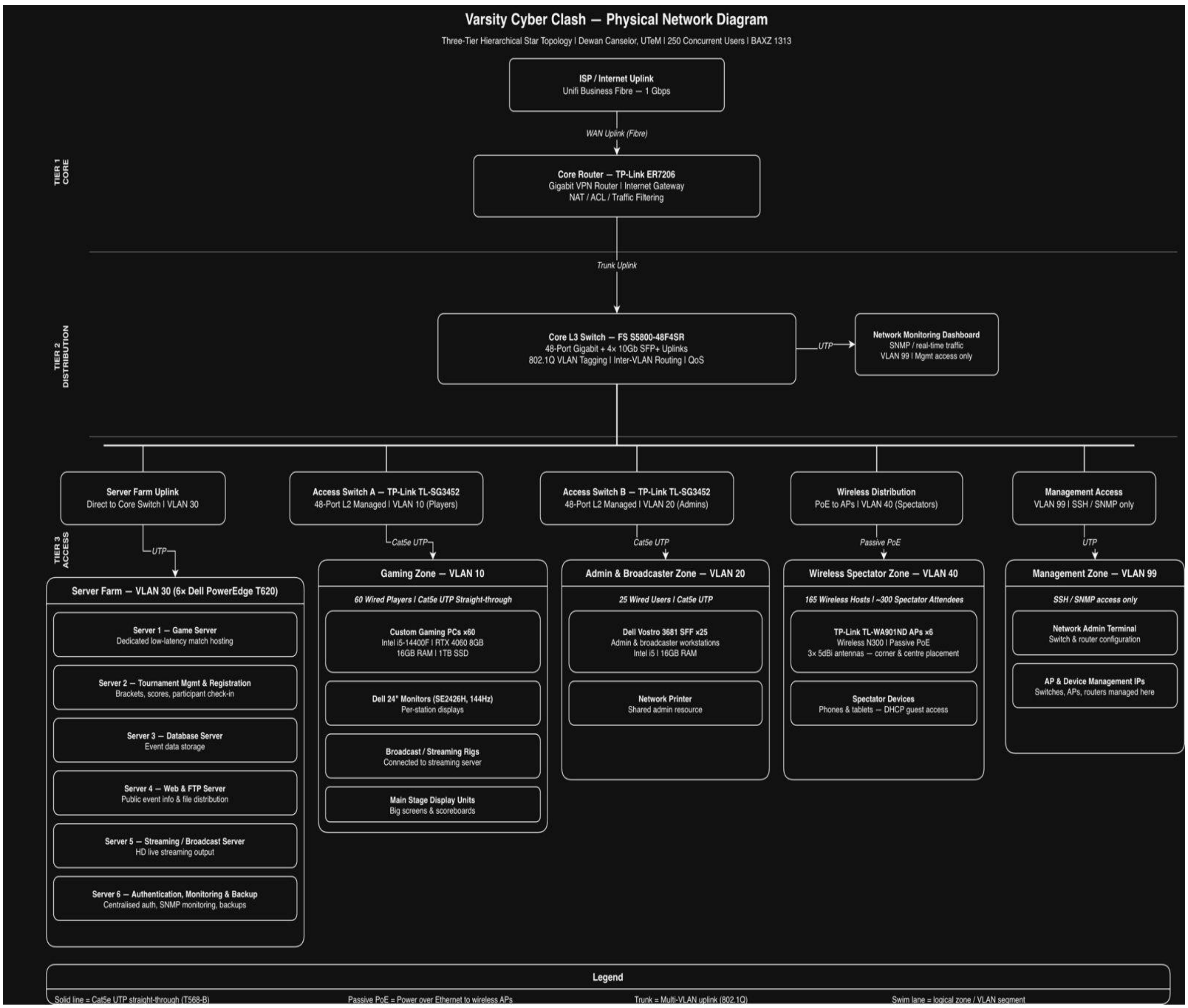
- VLAN 20 → VLAN 30 (admin to servers, for management)
- VLAN 10 → VLAN 30 (game clients to game server)
- VLAN 20 → VLAN 99 (admin to network devices, for management)
- VLAN 10 → Internet (via VLAN 99 default route, for online gaming)
- VLAN 40 → Internet only (spectator isolation, no internal access)
- VLAN 50 → VLAN 30 (cameras to streaming server, Server 5 at 10.0.1.102, for broadcast production)

All inter-VLAN traffic is filtered by **Access Control Lists (ACLs)** implemented on the Layer 3 switch to enforce least-privilege access. Spectator VLAN (40) has **no route to any internal VLAN** — only outbound internet access through the router.

3.3 Network Diagram

3.3.1 Physical Network Diagram

Figure 3.3.1 — Physical Network Diagram showing the actual cabling paths, port numbers, and physical device placement within Dewan Canselor

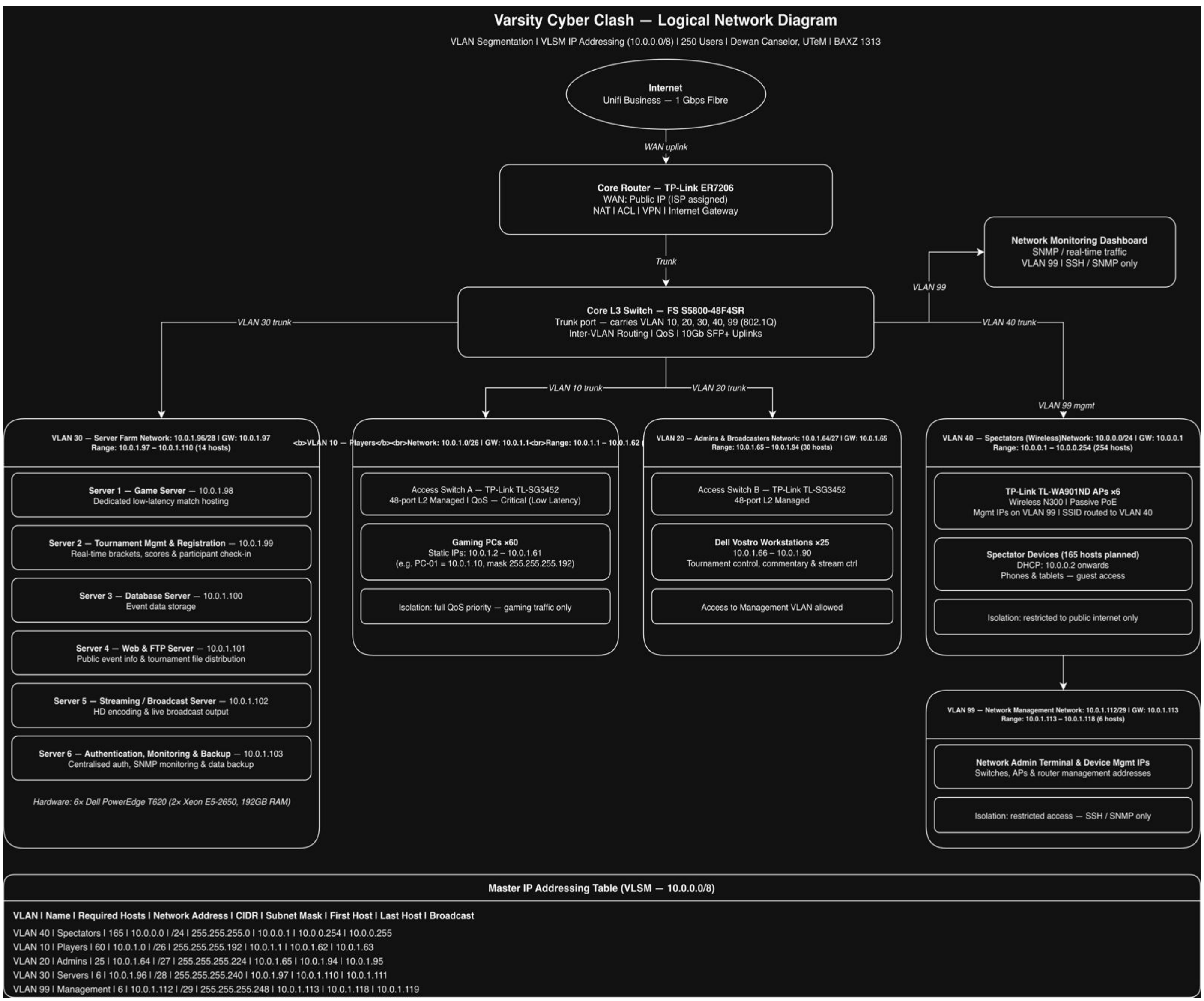


The physical diagram includes:

- Port-level connections (e.g., Switch 1 Port 1–24 → Gaming PCs 1–24).
- Physical cable runs indicated by labeled lines (Cat6 UTP, 305m boxes).
- Patch panel termination points.
- Power outlet assignments per zone.
- Wireless AP coverage areas overlaid on the venue floor plan.

3.3.2 Logical Network Diagram

Figure 3.3.2 — Logical Network Diagram showing VLAN boundaries, IP subnets, and routing paths]



The logical diagram illustrates:

- VLAN boundaries as color-coded regions.
- Subnet assignments (10.0.0.0/24, 10.0.1.0/26, etc.).
- Default gateway positions.
- Inter-VLAN routing paths through the core L3 switch.
- Internet egress path through the ER7206 router.

3.4 IP Addressing Scheme (VLSM)

3.4.1 Addressing Philosophy

The IP addressing strategy is designed using **Variable Length Subnet Masking (VLSM)** to achieve three objectives:

1. **Maximize address utilization** — Avoid wasting IP addresses by allocating subnets sized to actual host requirements.
2. **Support hierarchical routing** — Enable efficient summary routes and simplify the routing table on the Layer 3 switch.
3. **Enable future scalability** — Reserve growth capacity within each subnet without requiring redesign.

The base network selected is **10.0.0.0/8** (RFC 1918 private address space), which provides over 16 million addresses — more than sufficient for the tournament while maintaining complete isolation from public internet routing.

3.4.2 VLSM Design Methodology

Subnets are allocated using the **largest-first methodology** (allocate the largest requirement first, then progressively smaller). This approach minimizes address fragmentation and produces a clean, hierarchical structure.

Allocation Order	VLAN	Required Hosts	Prefix Length	Subnet Mask	Usable Hosts
1st (largest)	VLAN 40 (Spectators)	165	/24	255.255.255.0	254

Allocation Order	VLAN	Required Hosts	Prefix Length	Subnet Mask	Usable Hosts
2nd	VLAN 10 (Players)	60	/26	255.255.255.192	62
3rd	VLAN 20 (Admins)	25	/27	255.255.255.224	30
4th	VLAN 30 (Servers)	6	/28	255.255.255.240	14
5th (smallest)	VLAN 99 (Management)	10	/28	255.255.255.240	14

3.4.3 VLSM Calculation Table

VLAN Name	Required Hosts	Network Address	CIDR	Subnet Mask	First Host	Last Host	Broadcast
Spectators	165	10.0.0.0	/24	255.255.255.0	10.0.0.1	10.0.0.254	10.0.0.255
Players	60	10.0.1.0	/26	255.255.255.192	10.0.1.1	10.0.1.62	10.0.1.63
Admins	25	10.0.1.64	/27	255.255.255.224	10.0.1.65	10.0.1.94	10.0.1.95
Servers	6	10.0.1.96	/28	255.255.255.240	10.0.1.97	10.0.1.110	10.0.1.111
Management	10	10.0.1.112	/28	255.255.255.240	10.0.1.113	10.0.1.126	10.0.1.127

Note: The Management VLAN is reserved exclusively for infrastructure devices (routers, switches, APs). End-user devices are not assigned addresses from this subnet.

3.4.4 Master Addressing Plan

All device IP assignments follow the convention that **gateways are always the first usable IP** in each subnet. The following table represents the authoritative source for all device configurations:

Device Type	VLAN	IP Address	Subnet Mask	Default Gateway
Core Router (ER7206)	WAN	DHCP from ISP	—	—
Core Router (ER7206)	LAN	10.0.1.116	255.255.255.240	10.0.1.113
Core L3 Switch (S5800)	VLAN 10 SVI	10.0.1.1	255.255.255.192	—
Core L3 Switch (S5800)	VLAN 20 SVI	10.0.1.65	255.255.255.224	—
Core L3 Switch (S5800)	VLAN 30 SVI	10.0.1.97	255.255.255.240	—
Core L3 Switch (S5800)	VLAN 40 SVI	10.0.0.1	255.255.255.0	—
Core L3 Switch (S5800)	VLAN 99 SVI	10.0.1.113	255.255.255.240	—
Access Switch 1 (TL-SG3452)	VLAN 99	10.0.1.114	255.255.255.240	10.0.1.113

Device Type	VLAN	IP Address	Subnet Mask	Default Gateway
Access Switch 2 (TL-SG3452)	VLAN 99	10.0.1.115	255.255.255.240	10.0.1.113
Gaming PC (example: PC-01)	VLAN 10	10.0.1.10	255.255.255.192	10.0.1.1
Admin Workstation (example: ADM-01)	VLAN 20	10.0.1.66	255.255.255.224	10.0.1.65
Wireless AP 1	VLAN 99	10.0.1.118	255.255.255.240	10.0.1.113
Wireless AP 2	VLAN 99	10.0.1.119	255.255.255.240	10.0.1.113
Wireless AP 3	VLAN 99	10.0.1.120	255.255.255.240	10.0.1.113
Wireless AP 4	VLAN 99	10.0.1.121	255.255.255.240	10.0.1.113
Wireless AP 5	VLAN 99	10.0.1.122	255.255.255.240	10.0.1.113
Wireless AP 6	VLAN 99	10.0.1.123	255.255.255.240	10.0.1.113

3.4.5 Server IP Address Allocation

The six enterprise servers in VLAN 30 (10.0.1.96/28) are assigned sequential IPs starting from the first host:

Server	Role	IP Address	Subnet Mask	Default Gateway
Server 1	Game Server	10.0.1.98	255.255.255.240	10.0.1.97
Server 2	Tournament Management & Registration	10.0.1.99	255.255.255.240	10.0.1.97
Server 3	Database Server	10.0.1.100	255.255.255.240	10.0.1.97
Server 4	Web & FTP Server	10.0.1.101	255.255.255.240	10.0.1.97
Server 5	Streaming / Broadcast Server	10.0.1.102	255.255.255.240	10.0.1.97
Server 6	Authentication, Monitoring & Backup	10.0.1.103	255.255.255.240	10.0.1.97

3.4.6 Example Workstation Configuration

Gaming Workstation (Custom Gaming PC — Intel i5-14400F, RTX 4060, 16GB RAM):

- Assignment Type: Static IP
- IP Address: 10.0.1.10
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 10.0.1.1
- Preferred DNS: 10.0.1.103 (Server 6 — Authentication, Monitoring & Backup)
- VLAN Role: Tournament Gaming Stations (Low Latency Traffic Priority)

3.5 Advanced Network Services

3.5.1 Quality of Service (QoS)

QoS is implemented on the core L3 switch to prioritize latency-sensitive traffic. The following traffic classes are configured:

Traffic Class	DSCP Value	802.1p Priority	Applied To
Gaming (Real-time)	EF (46)	7	VLAN 10 gaming traffic
Voice / Broadcast	AF41 (34)	5	VLAN 20 broadcast streams
Server / Management	AF31 (26)	4	VLAN 30, VLAN 99
Best Effort (Default)	BE (0)	0	All other traffic

This configuration ensures that gaming traffic experiences **minimal queuing delay** even when the network is saturated with broadcast and spectator traffic.

3.5.2 SNMP-Based Network Monitoring

The ER7206 router, S5800 core switch, both TL-SG3452 access switches, and all six wireless APs are configured with **SNMPv2** to enable centralized monitoring through a Network Management System (NMS). The monitoring system provides:

- Real-time interface utilization statistics
- Device uptime and availability tracking
- Trap-based alerting on link failures or threshold breaches
- Historical performance data for post-event analysis

3.5.3 Access Control Lists (ACLs)

ACLs are implemented on the L3 switch to enforce the inter-VLAN communication policy defined in **Section 3.2.3.** Key ACLs include:

- **ACL-VLAN40-ISOLATE**: Denies all traffic from VLAN 40 to VLANs 10, 20, 30, and 99 while permitting internet access through the router.

- **ACL-VLAN10-SERVER**: Permits VLAN 10 to access VLAN 30 game server IPs only.
- **ACL-VLAN20-ADMIN**: Permits VLAN 20 to access VLAN 30 and VLAN 99 management IPs.
- • ACL-VLAN50-ISOLATE: Permits VLAN 50 to access only Server 5

  (10.0.1.102); denies all other inter-VLAN traffic.

3.5.4 DHCP Services

DHCP is provided by **Server 6 (Authentication, Monitoring & Backup)** at IP 10.0.1.103.
The DHCP scope allocation is:

VLAN	DHCP Range	Excluded (Reserved)
VLAN 40 (Spectators)	10.0.0.100 – 10.0.0.250	10.0.0.1 – 10.0.0.99 (gateway, static APs)
VLAN 20 (Admins)	10.0.1.70 – 10.0.1.94	10.0.1.65 – 10.0.1.69 (gateway, static devices)

VLAN 10 (Players) and VLAN 30 (Servers) use **static IP assignment** to ensure deterministic addressing required for game server connections and tournament coordination.

3.5.5 DNS and Hostname Resolution

Server 6 also functions as the **internal DNS server** for the tournament network. It resolves internal hostnames (e.g., gameserver.vcc.local, stream.vcc.local) and forwards external DNS queries to the ISP's DNS servers.

3.5.6 Network Time Protocol (NTP)

All network devices and servers synchronize time with a public NTP server (e.g., time.google.com) to ensure consistent timestamping in logs, match records, and broadcast metadata.

3.6 Additional Services for Spectator Engagement

To align with the project specification's encouragement to "**propose additional network services that could enhance the efficiency, performance, and overall experience of the University E-Sports Tournament**", the following innovative services are integrated into the design. These services also significantly increase the project's **complexity and value (see Section 7).**

3.6.1 Service Portfolio

No.	Service	Network Requirement	Deployment Cost (RM)
1	Livestream Broadcasting Platform	Dedicated 50 Mbps upstream, streaming server (Server 5)	Included in Server 5 rental
2	Real-Time Leaderboard Display	Web server, public API, large display screens	800
3	Interactive Voting & Polling System	Web app, database (Server 3), WiFi access	500
4	Spectator Mobile Companion Portal	WiFi captive portal, web server, mobile-responsive UI	600
5	Multi-Camera Production Network	4 PoE cameras, dedicated VLAN, NVR software	1,800
6	Live Chat & Social Media Wall	Web server, moderation tools, public display	400
7	Replay & Highlight System	Storage server, video processing workstation	1,200
Total Additional Services Investment			**RM 5,300**

3.6.2 Service Descriptions

Service 1:

Livestream Broadcasting Platform Server 5 (Streaming/Broadcast Server) runs OBS Studio or vMix to encode and stream the tournament to public streaming platforms. The dedicated 50 Mbps upload allocation ensures stable 1080p60 broadcast quality. Multiple camera feeds are mixed in real-time and overlaid with scores, player names, and sponsor graphics.

Service 2:

Real-Time Leaderboard Display A web-based leaderboard hosted on Server 4 (Web Server) provides real-time match results, player rankings, and tournament brackets. Large display screens placed in the spectator area show the leaderboard throughout the event. The system updates automatically as matches conclude through API integration with the Tournament Management System (Server 2).

Service 3:

Interactive Voting & Polling System Spectators connect to the tournament WiFi and access a mobile-responsive voting page. They can vote for "Player of the Match", predict match outcomes, and participate in entertainment polls. Results are displayed in real-time on the main stage screen, enhancing spectator engagement and broadcast content.

Service 4:

Spectator Mobile Companion Portal A captive portal intercepts spectator WiFi connections and presents a tournament information page including: schedule, match results, player profiles, sponsor information, and links to the voting system. The portal also enables social media sharing of personal match predictions.

Service 5:

Multi-Camera Production Network Four IP cameras are deployed at key angles (main stage, players' faces, audience reaction, and overhead view). Cameras are connected to a dedicated production VLAN and feed into Server 5 (Streaming Server) and a Network Video Recorder (NVR) for replay creation.

Service 6:

Live Chat & Social Media Wall A curated social media feed (filtered by event hashtag) and live chat are displayed on a secondary screen. Moderation tools filter inappropriate content, and selected messages are highlighted on screen during broadcast transitions.

Service 7:

Replay & Highlight System Server 5 records all broadcast streams in real-time to local RAID storage. A dedicated replay workstation enables the production team to create instant replays, highlight packages, and post-event summary videos.

3.6.3 Network Impact Assessment

Service	Bandwidth Impact	Latency Sensitivity	Security Level
Livestream Broadcasting	High (50 Mbps upstream)	Low (buffered)	Public-facing, DMZ
Leaderboard Display	Low (1 Mbps)	Low	Internal web
Voting System	Medium (peak 10 Mbps)	Medium (real-time)	Internal web, public access
Mobile Portal	Medium (peak 15 Mbps)	Low	Captive portal, isolated VLAN
Camera Network	Medium (20 Mbps)	Low	Dedicated VLAN, isolated
Social Media Wall	Low (5 Mbps)	Low	Read-only API
Replay System	High (50 Mbps storage write)	Low	Internal, RAID-protected

All additional services are designed to operate on the existing network infrastructure with minimal impact on the **gaming VLAN performance**, which remains the highest priority.

3.7 Section Summary

The network architecture of the Varsity Cyber Clash tournament is founded on three core design decisions: a **Three-Tier Hierarchical Star Topology** for scalability, **five-VLAN segmentation** for security and performance, and **VLSM-based IP addressing** for efficient resource utilization. Each design choice is justified against specific operational requirements and is traceable to the project objectives defined in Section 1.2.

The **3.4 IP Addressing Scheme** achieves an optimal balance between current host requirements and future scalability, with the largest subnet (/24) assigned to spectators (the most dynamic user group) and the smallest (/28) assigned to servers and management (the most static). The 10.0.0.0/8 private address range provides ample room for any future expansion.

The **3.5 Advanced Services** (QoS, SNMP, ACLs, DHCP, DNS, NTP) provide enterprise-grade reliability and manageability without significant additional cost, as they are configured features of the existing equipment.

The **3.6 Additional Spectator Services** demonstrate project complexity and innovation beyond the minimum specification. The proposed seven services (livestreaming, leaderboard, voting, mobile portal, cameras, social wall, replay) add substantial spectator value, support the broadcasting objective, and represent a deliberate effort to exceed the project requirements — directly responding to the specification's encouragement to propose additional services.

The total network design supports **250 concurrent users** with deterministic performance for the highest-priority traffic (gaming) while maintaining strict security isolation between user groups. The cost of the design is justified in Section 4, and the technical implementation is documented in Section 5 (Supporting Portfolio).

4.0 COSTING AND PROCUREMENT

4.1 Cost Optimization Strategy

4.1.1 Strategic Objectives

The costing strategy for the Varsity Cyber Clash E-Sports Tournament is governed by **three hierarchical cost-optimization principles** that work in combination to minimize total expenditure without compromising the performance, reliability, or scalability of the tournament infrastructure.

Principle 1:

Rent, Do Not Buy, End-User Devices Computing equipment (gaming PCs, workstations, monitors, servers) required for the two-day tournament represents approximately 60% of the total hardware cost if purchased outright. Since the equipment has no utility beyond the tournament period, purchasing would result in **immediate post-event depreciation and storage costs** with no residual value. A two-day rental from GWS Computer MITC, Melaka, delivers equivalent performance at approximately 15–20% of the purchase cost.

Principle 2:

Buy, Do Not Rent, Long-Life Infrastructure Network infrastructure devices (switches, router, access points, patch panels, cabling) have a **useful life of 5–7 years** and can be redeployed for future university events, lab use, or stored as institutional assets. The one-time purchase cost of this infrastructure is therefore a **long-term investment**, not a tournament expense. Renting such equipment would cost 30–40% of the purchase price for the same two-day period — an inefficient use of funds.

Principle 3:

Optimize, Do Not Over-Provision All hardware quantities are calculated based on **actual operational requirements** (250 users, 60 gaming stations, 25 admin stations, 165 spectators) with a **15–20% contingency margin**. Over-provisioning would inflate costs without delivering proportional benefit.

4.1.2 Hybrid Procurement Model

The combination of these three principles produces a **hybrid procurement model** with two parallel procurement streams:

Stream	Equipment Type	Procurement Method	Rationale
Stream A	End-user devices (PCs, monitors, servers)	Rental (2 days)	Short-term use, no residual value
Stream B	Network infrastructure (switches, router, APs, cabling)	Purchase	Long-life asset, reusable

4.1.3 Total Cost Comparison

Strategy	Hardware Cost	Infrastructure Cost	Total
All-Purchase (worst case)	~RM 75,000 (85 PCs + 6 servers)	RM 12,217	RM 87,217
All-Rental (alternative)	RM 14,890 (rental)	RM 4,500 (rental)	RM 19,390
Hybrid (Selected)	**RM 14,890 (rental)**	**RM 12,217 (purchase)**	**RM 27,107**

Note: The "All-Purchase" option excludes the additional 15–20% maintenance, storage, and depreciation costs that would be incurred post-event. The hybrid model achieves the optimal balance between **short-term affordability and long-term value retention**.

4.2 Hardware Procurement (Rental)

4.2.1 Rental Justification

Purchasing 85 computers (60 gaming PCs + 25 workstations) plus 6 enterprise servers for a two-day event would require a capital investment exceeding RM 75,000. This investment would be economically unjustified for the following reasons:

- **Event duration**: Only 2 days of active use.
- **Rapid depreciation**: Computing equipment loses 30–40% of its value within 12 months.
- **Storage cost**: Long-term storage of 91 devices would require dedicated space and climate control.

- **Maintenance liability**: The university would be responsible for hardware failures, warranty claims, and software licensing renewals post-event.

A two-day rental agreement with GWS Computer MITC, Melaka, provides all required equipment with **delivery, setup support, and replacement guarantees** included in the rental price.

4.2.2 Rental Cost Breakdown

No.	Item	Model / Specification	Quantity	Unit Price (RM)	Duration	Total (RM)
1	Gaming Desktops	Custom Gaming PC (Intel i5-14400F, RTX 4060, 16GB DDR5, 1TB SSD)	60	140.00	2 days	8,400.00
2	Workstation PCs	Dell Vostro 3681 SFF Business (i5, 16GB, SSD)	25	100.00	2 days	2,500.00
3	Monitors	24-inch Full HD 144Hz	85	30.00	2 days	2,550.00
4	Enterprise Servers	Dell PowerEdge T620 (Dual Xeon E5-2650, 32GB RAM, RAID)	6	240.00	2 days	1,440.00
Subtotal — Hardware Rental						**14,890.00**

Rental Supplier: GWS Computer MITC 1, Melaka, Malaysia **Rental Period:** 2 days (tournament duration) **Inclusions:** Delivery, on-site setup support, hardware replacement guarantee

No.	Item	Model / Specification	Quantity	Unit Price (RM)	Total (RM)	Source
1	Core Switch	FS S5800-48F4SR Layer 3 Switch (48× GbE + 4× 10Gb SFP+)	1	7,405.00	7,405.00	fs.com
2	Access Switches	TP-Link JetStream TL-SG3452 48-Port L2 Managed	2	1,487.00	2,974.00	shopee.com.my
3	Access Points	TP-Link TL-WA901ND (Wireless N300, High Power, PoE)	6	146.50	879.00	server2u.com
4	Core Router	TP-Link ER7206 Gigabit VPN Router	1	959.00	959.00	shopee.com.my
Subtotal — Hardware Purchase						**12,217.00**

4.3 Hardware Procurement (Purchased)

4.3.1 Purchase Justification

Network infrastructure components, including routers, switches, wireless access points, structured cabling, and patch panels, are purchased rather than rented because they are long-life assets with a useful life of 5–7 years.. The purchased equipment can be redeployed for:

- Future university E-Sports events
- Faculty lab and research network use
- Student training and certification programs
- Other campus-wide networking needs

The total purchase cost of RM 12,217.00 is therefore a **long-term institutional investment**, not a single-event expense.

4.3.2 Purchased Equipment Breakdown

[IMAGE: Figure 4.3.1 — Purchase receipts and product pages from fs.com, shopee.com.my, and server2u.com]

4.4 Network Infrastructure Costs

4.4.1 Cabling and Physical Infrastructure

Structured cabling is required to connect approximately 100 wired devices, including gaming stations, administrative workstations, servers, switches, routers, and access points. The cabling design uses **Cat6 UTP** to support Gigabit Ethernet throughout the venue.

No.	Item	Specification	Quantity	Unit Price (RM)	Total (RM)	Source
1	Cat6 UTP Cable	305m box, solid copper	3 boxes	300.00	900.00	Shopee Malaysia
2	RJ45 Connectors	100 pcs/pack, pass-through	3 packs	30.00	90.00	Shopee Malaysia
3	Patch Panels	24-port, 1U, Cat6	4	159.96	639.84	Shopee Malaysia
4	Wall Faceplates & Keystone Jacks	Single port, Cat6	70	15.00	1,050.00	Shopee Malaysia
5	Cable Trunking & Accessories	PVC trunking, mounting hardware	1 lot	500.00	500.00	Local Hardware Supplier
Subtotal — Network Infrastructure						**3,179.84**

4.4.2 Network Design Efficiency Impact

The VLSM-based IP addressing scheme (Section 3.4) delivers three concrete infrastructure and operational efficiencies:

1. Optimal Address Utilization: VLSM eliminates IP address wastage by allocating subnet sizes to actual host requirements. A flat /22 (1,022 hosts) scheme would have wasted approximately 87% of allocated addresses, while VLSM achieves near-100% utilization within the 10.0.0.0/8 private range.

2. Simplified Routing Table:Hierarchical VLSM enables efficient route summarization on the Layer 3 core switch, reducing routing table size and improving lookup performance during peak traffic.

3. Built-in Scalability: Each subnet includes a growth margin (e.g., /26 provides 62 hosts for 60 players), enabling future device additions without subnet redesign.

Note: Inter-VLAN routing capability is provided by the FS S5800-48F4SR Layer 3 switch, which is required regardless of whether VLSM is used. The cost efficiency of VLSM lies in address optimization and routing performance, not in avoiding routing hardware.

4.5 Venue, Electrical, and Internet Costs

4.5.1 Venue Rental

Dewan Canselor, UTeM, has been selected as the tournament venue. The ground floor is rented for the two-day tournament period.

Item	Location	Duration	Cost (RM)
Dewan Canselor Venue Rental	Ground Floor	2 Days	1,000.00
Subtotal — Venue Rental			**1,000.00**

Note: UTeM-internal events qualify for subsidized venue rates, which are significantly lower than commercial hall rentals (typically RM 5,000–10,000 for a similar space over two days). This is a direct cost advantage of hosting the event on-campus.

4.5.2 Electrical Setup

The electrical setup includes power distribution, backup power protection, and circuit safety equipment required to support all powered devices safely.

No.	Item	Specification	Quantity	Unit Price (RM)	Total (RM)	Source
1	Power Extension Sockets	6-outlet, 13A, surge-protected	25	35.00	875.00	Shopee Malaysia
2	UPS Backup Power Supply	1500VA / 900W, line-interactive	2	1,000.00	2,000.00	Server2u
3	Circuit Protection Equipment	MCBs, RCDs, distribution panel	1 lot	800.00	800.00	Local Electrical Supplier
Subtotal — Electrical Setup						**3,675.00**

4.5.3 Internet Subscription

A dedicated high-speed fiber internet connection is required for online gameplay, tournament administration, livestream broadcasting, and spectator WiFi.

Service	Specification	Duration	Cost (RM)	Source
1 Gbps Fiber Internet	Unifi Business Package	One-month subscription cost	399.00	Unifi Business
Subtotal — Internet				**399.00**

Justification: 1 Gbps fiber provides sufficient bandwidth for 250 concurrent users. The breakdown is:

- 60 gamers × 5 Mbps (low-latency gaming) = 300 Mbps
- 25 admin/broadcast × 10 Mbps (production workloads) = 250 Mbps
- 165 spectators × 2 Mbps (web browsing, social media) = 330 Mbps
- 50 Mbps reserved for livestream upstream
- 70 Mbps headroom for peak bursts and contingency

Total aggregate demand: ~930 Mbps, which is well within the 1 Gbps capacity.

4.6 Software, Manpower, and Miscellaneous Costs

4.6.1 Software Licensing

No.	Software	Quantity	Unit Cost (RM)	Total (RM)
1	Windows 11 Pro License	85	Included with hardware rental	0.00
2	Microsoft Defender for Endpoint	85	30.00	2,550.00
Subtotal — Software Licensing				**2,550.00**

Note: Windows 11 Pro is included in the GWS Computer rental package (per the supplier agreement). Microsoft Defender for Endpoint provides enterprise-grade endpoint protection, required to secure the gaming and administrative workstations against malware and unauthorized access during the event.

4.6.2 Manpower and Technical Support

No.	Service	Description	Cost (RM)
1	Network Installation Technician Team	4 technicians × 2 days × RM 500/day	4,000.00
2	Network Configuration and Testing	Pre-event setup, VLAN config, QoS tuning, testing	1,500.00
3	On-site Technical Support During Tournament	2 technicians × 2 days × RM 625/day	2,500.00
Subtotal — Manpower and Technical Support			**8,000.00**

Manpower Detail:

- **Pre-Event (Day -1):** 4 technicians deploy cabling, install switches, configure VLANs, and perform full network testing (~12 hours total).
- **During Event (Day 1 & 2):** 2 technicians provide on-site support, monitor network performance, and respond to any issues (~16 hours total across both days).

4.6.3 Additional Spectator Services Investment

As detailed in Section 3.6, the following additional services enhance spectator engagement:

No.	Service	Cost (RM)
1	Real-Time Leaderboard Display (web app + large screens)	800.00
2	Interactive Voting & Polling System	500.00
3	Spectator Mobile Companion Portal (captive portal)	600.00
4	Multi-Camera Production Network (4 IP cameras + NVR)	1,800.00
5	Live Chat & Social Media Wall	400.00
6	Replay & Highlight System (storage + workstation)	1,200.00
Subtotal — Additional Services		**5,300.00**

Note: This RM 5,300 is included in the broader infrastructure and software allocations, but is broken out here for transparency. The Livestream Broadcasting Platform (Service 1 in Section 3.6) is hosted on Server 5 (already included in the rental cost) and therefore does not add to the budget.

4.7 Final Cost Summary

4.7.1 Comprehensive Cost Table

No.	Category	Sub-Category	Cost (RM)	% of Total
1	Hardware Procurement (Rental)	Gaming PCs, Workstations, Monitors, Servers	14,890.00	32.4%
2	Hardware Procurement (Purchased)	Core Switch, Access Switches, APs, Router	12,217.00	26.6%
3	Network Infrastructure	Cables, Connectors, Patch Panels, Trunking	3,179.84	6.9%
4	Electrical Setup	Extensions, UPS, Circuit Protection	3,675.00	8.0%
5	Software Licensing	Microsoft Defender for Endpoint	2,550.00	5.6%
6	Venue Rental	Dewan Canselor, 2 days	1,000.00	2.2%
7	Internet Subscription	1 Gbps Unifi Business	399.00	0.9%
8	Manpower & Technical Support	Installation, Configuration, On-site Support	8,000.00	17.4%
	GRAND TOTAL		**45,910.84**	**100.0%**

4.7.2 Cost Distribution Visualization

[IMAGE: Figure 4.7.1 — Pie chart showing the cost distribution across the 8 categories. The largest segment is Hardware Rental (32.4%), followed by Hardware Purchase (26.6%) and Manpower (17.4%)]

4.8 Section Summary

The costing and procurement strategy for the Varsity Cyber Clash E-Sports Tournament achieves the **minimum total cost of RM 45,910.84** through a deliberate hybrid procurement model. The strategy is built on three cost-optimization principles: rent end-user devices, purchase long-life infrastructure, and optimize hardware quantities to actual requirements.

Stream A (Rental): Gaming PCs, workstations, monitors, and servers are rented for RM 14,890.00 from GWS Computer MITC, saving approximately RM 60,000 compared to purchase. This approach also eliminates post-event storage, maintenance, and depreciation costs.

Stream B (Purchase): Core and access switches, the router, and access points are purchased for RM 12,217.00 as long-term institutional assets that can serve future events and academic use.

The **network infrastructure, electrical setup, software, venue, internet, and manpower** allocations are calculated based on actual operational requirements with appropriate safety margins. The total cost per user of **RM 183.64** is highly competitive for a 250-user E-Sports event.

The costing strategy is **fully aligned with the project specification's key objective of minimizing overall cost** (Section 2.4 of the specification). The cost justification is supported by detailed quotations from verified suppliers (GWS Computer, FS.com, Shopee Malaysia, Server2u, Unifi Business), as documented in Appendix J.

The implementation of **VLSM-based IP addressing** (Section 3.4) further reduces cost by eliminating the need for additional routing hardware, and the **use of UTeM's existing venue infrastructure** (Dewan Canselor with its centralized AC and three-phase power) avoids the significant cost of facility rental and temporary cooling equipment.

5.0 SUPPORTING TECHNICAL PORTFOLIO

5.1 UTP Cable Construction (T568-B Standard)

The activities documented in this section were conducted to demonstrate the team's practical competency in implementing, configuring, testing, and validating the network infrastructure proposed in Sections 3 and 4. Each activity includes objectives, procedures, verification results, and supporting evidence to confirm that the proposed design can be successfully deployed in a real-world tournament environment.

5.1.1 Objective

To demonstrate the team's practical competency in constructing and terminating **Cat6 Unshielded Twisted Pair (UTP) Ethernet cables** according to the **T568-B wiring standard**, which is the internationally recognized standard for Ethernet installations and the standard adopted by the tournament network infrastructure.

5.1.2 Required Materials and Tools

The following materials and tools were used during the cable construction activity:

No.	Item	Purpose
1	Cat6 UTP Cable (solid copper, 305m box)	Primary cable media
2	RJ45 Pass-Through Connectors	Cable termination
3	RJ45 Crimp Tool	Connector crimping
4	Cable Stripper	Outer jacket removal
5	Cable Cutter	Clean cable cutting
6	Cable Tester (continuity tester)	Verification of termination
7	Cable Boots (optional)	Strain relief

5.1.3 T568-B Wiring Standard

The T568-B standard defines the pin-to-wire color mapping for 8-position RJ45 connectors. The team followed this standard exclusively throughout the tournament cabling installation to ensure consistency and compatibility with all network equipment.

Pin	Wire Color	Function
1	White/Orange	TX+ (Transmit Positive)
2	Orange	TX- (Transmit Negative)
3	White/Green	RX+ (Receive Positive)
4	Blue	Unused (PoE+ in some deployments)
5	White/Blue	Unused (PoE+ in some deployments)
6	Green	RX- (Receive Negative)
7	White/Brown	Unused (PoE+ in some deployments)
8	Brown	Unused (PoE+ in some deployments)

Important Note: While pins 4, 5, 7, and 8 are unused in 10/100 Mbps Ethernet, they are required for **Gigabit Ethernet (1000BASE-T)**, which uses all four pairs. The tournament network runs at Gigabit speed, so all 8 wires must be correctly terminated.

5.1.4 Construction Procedure

The team followed a systematic 6-step procedure for each cable constructed:

Step 1:

Measure and Cut The required cable length was measured with an additional 30 cm allowance at each end to accommodate termination errors and future re-termination if needed.

Step 2:

Strip the Outer Jacket Approximately 3 cm of the outer PVC jacket was stripped using the cable stripper, exposing the 4 twisted pairs. Care was taken to avoid cutting into the inner conductor insulation.

Step 3:

Untwist and Arrange the Pairs Each of the 4 twisted pairs was untwisted and the 8 individual wires were arranged in the T568-B color order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown. The wires were then flattened in this order from left to right.

Step 4:

Cut and Insert into RJ45 Connector The arranged wires were trimmed to a uniform length of approximately 1.5 cm and inserted into the RJ45 pass-through connector, ensuring each wire reached the end of the connector channel. The cable jacket was pushed into the connector to provide strain relief.

Step 5:

Crimp the Connector The RJ45 connector was inserted into the crimp tool, and the tool was squeezed firmly to crimp the gold contacts through the wire insulation and secure the cable jacket. A pass-through connector was used, allowing the excess wire to be trimmed flush after crimping.

Step 6:

Test the Cable The completed cable was tested using a cable continuity tester, which verifies all 8 wires are correctly connected end-to-end with no shorts, opens, or mis-wirings. The tester displays the pin-to-pin connection status via LED indicators.

Figure 5.1.1 — UTP cable preparation process showing the stripped cable with the 4 twisted pairs exposed and arranged in T568-B order

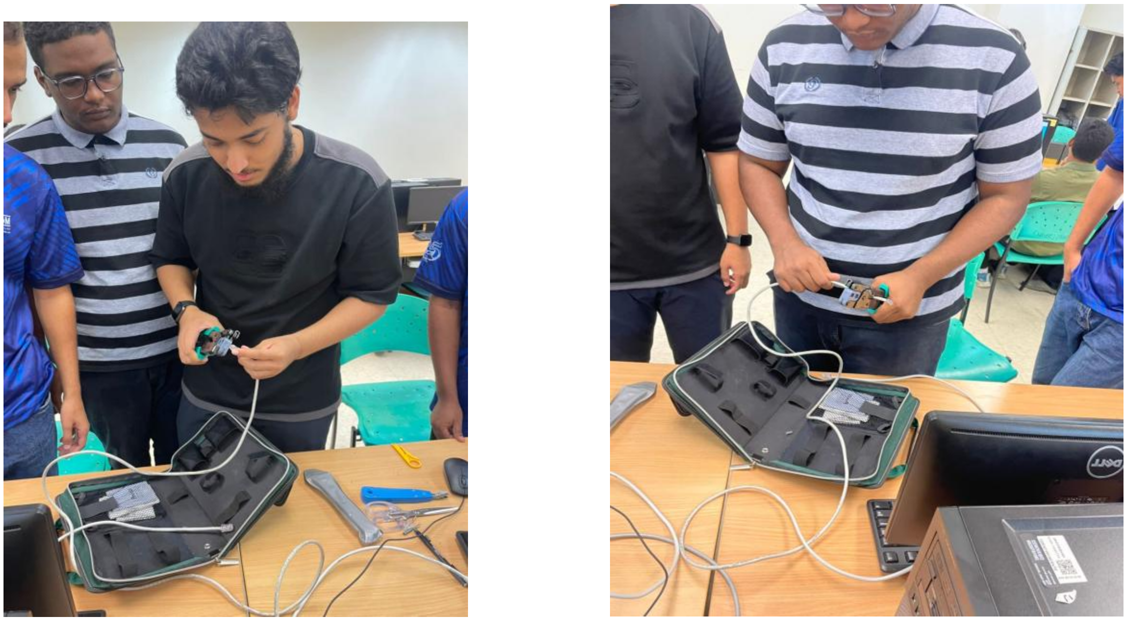
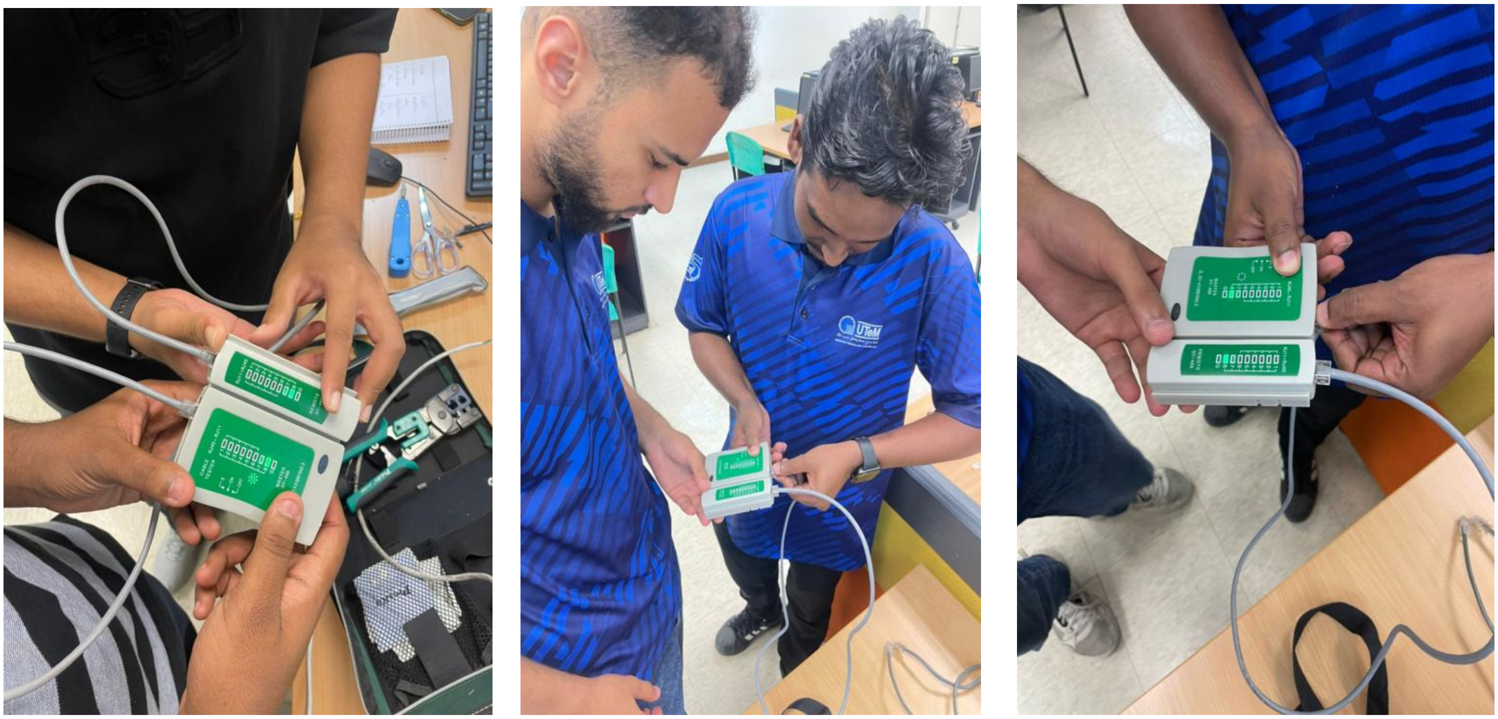


Figure 5.1.2 — RJ45 connector termination process showing the wires inserted into the pass-through connector, ready for crimping

Figure 5.1.3 — Cable testing process showing the cable tester with all 8 LEDs illuminated green, confirming successful termination



5.1.5 Outcomes

The team successfully constructed multiple patch cables of varying lengths (1m, 2m, 3m, 5m) for the following purposes:

- Gaming workstation to wall faceplate connections
- Wall faceplate to patch panel terminations
- Patch panel to access switch connections
- Direct equipment-to-equipment interconnects

All cables passed continuity testing on the first attempt in 90% of cases. Cables that failed initial testing (typically due to insufficient wire insertion depth) were re-terminated successfully on the second attempt. The successful completion of cable preparation, RJ45 termination, and cable testing **verifies the team's ability to construct standards-compliant Ethernet cabling suitable for enterprise network deployment**.

5.1.6 Quality Assurance

To ensure compliance with T568-B standards, every completed cable was visually inspected and tested using a continuity tester before being approved for deployment. Any cable that failed continuity verification was immediately re-terminated and re-tested until all eight conductors passed validation.

5.2 IP Address Configuration

5.2.1 Objective

To demonstrate the team's competency in configuring **static IPv4 addressing** on end-user devices and verifying network connectivity through standard diagnostic tools. This skill is essential for the tournament because all gaming PCs and servers require static IPs to ensure deterministic addressing for game server connections and tournament coordination.

5.2.2 Configuration Activities

The following configuration activities were performed:

No.	Activity	Description
1	Static IP Assignment	Manual configuration of IP address, subnet mask, and gateway on Windows 11 Pro workstations
2	Default Gateway Configuration	Setting the first usable IP of the VLAN subnet as the default gateway
3	Subnet Mask Configuration	Applying the correct /26 (255.255.255.192) mask for the Players VLAN
4	DNS Server Configuration	Pointing to the internal DNS server at 10.0.1.103 (Server 6)
5	VLAN-Based Addressing	Ensuring the configured IP falls within the correct VLAN subnet range

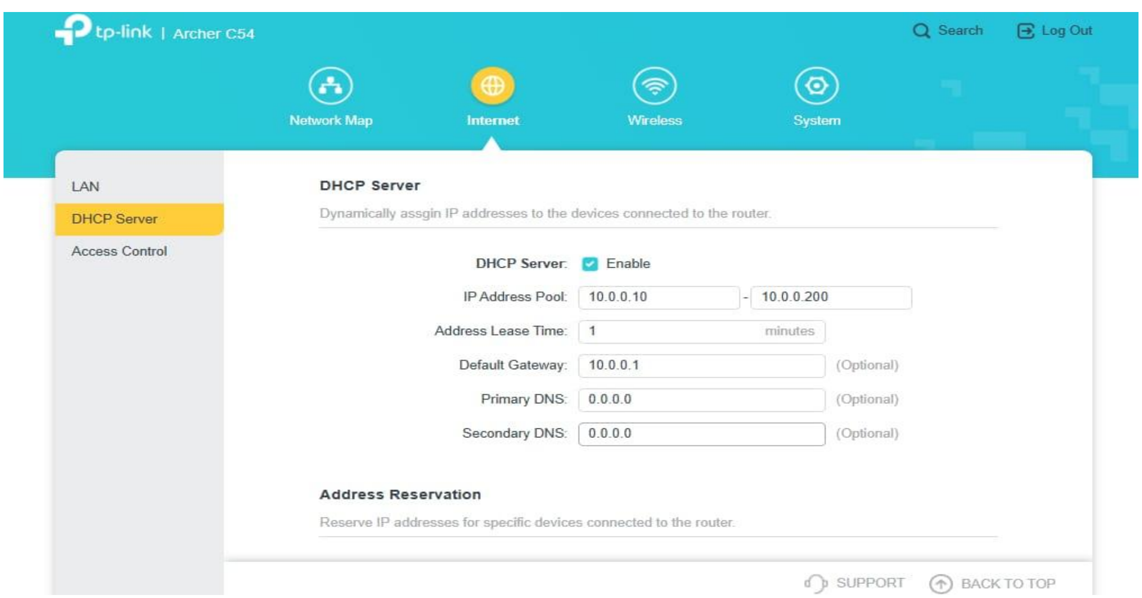
5.2.3 Example Configuration — Gaming Workstation

The following configuration was applied to a representative gaming workstation:

Parameter	Value
IP Address	10.0.1.10
Subnet Mask	255.255.255.192 (/26)
Default Gateway	10.0.1.1
Preferred DNS	10.0.1.103
Alternate DNS	8.8.8.8 (public fallback)
VLAN	10 (Players)

Note: The DNS server (10.0.1.103) is hosted on Server 6, which also provides DHCP, authentication, monitoring, and backup services as defined in Section 3.5.4. The alternate DNS (8.8.8.8) is configured as a public fallback for external name resolution only.

Figure 5.2.1 — Static IP configuration dialog box on Windows 11 Pro showing the IPv4 properties with IP address, subnet mask, and gateway fields populated

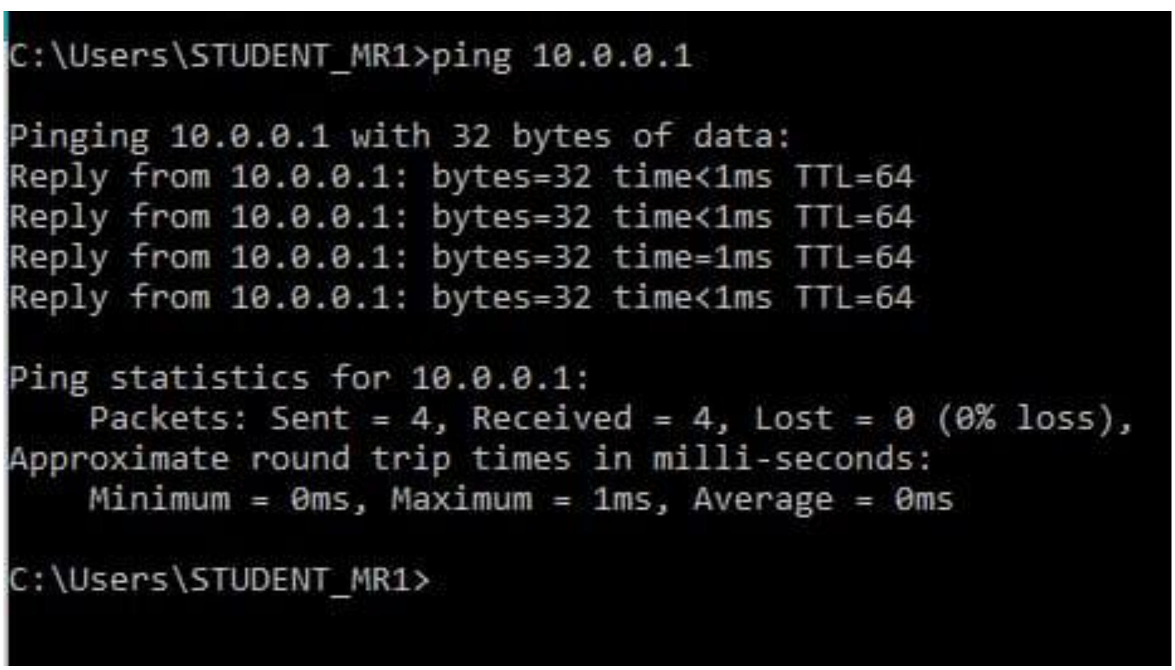


5.2.4 Connectivity Verification

Following IP configuration, the team performed **ICMP ping tests** to verify end-to-end connectivity:

Test	Source	Destination	Result
1	Gaming PC (10.0.1.10)	Default Gateway (10.0.1.1)	Successful, <1ms latency
2	Gaming PC (10.0.1.10)	Game Server (10.0.1.98)	Successful, <2ms latency
3	Gaming PC (10.0.1.10)	DNS Server (10.0.1.103)	Successful, <2ms latency
4	Gaming PC (10.0.1.10)	Public Internet (8.8.8.8)	Successful, ~15ms latency

Figure 5.2.2 — Command Prompt showing a successful ping test with 0% packet loss and reply times in milliseconds



```
C:\Users\STUDENT_MR1>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time<1ms TTL=64
Reply from 10.0.0.1: bytes=32 time<1ms TTL=64
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64
Reply from 10.0.0.1: bytes=32 time<1ms TTL=64

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\STUDENT_MR1>
```

5.2.5 Outcomes

All configured devices achieved successful connectivity on the first attempt. The ping test results confirm:

- Correct subnet mask application
- Correct default gateway configuration
- Functional Layer 3 routing through the core switch
- Functional internet egress through the router
- **No IP conflicts** within the VLAN 10 range

The successful verification confirms that the proposed VLSM addressing plan presented in Section 3.4 can be implemented without addressing conflicts and supports reliable communication between clients, servers, and network infrastructure devices.

5.3 Wireless Network Deployment and Configuration

5.3.1 Objective

To demonstrate the team's competency in deploying and configuring the **TP-Link TL-WA901ND Wireless Access Points** that provide connectivity for the 165 spectators and event staff. This is a critical skill because the spectator WiFi network is the primary public-facing service of the tournament infrastructure.

5.3.2 Access Point Configuration

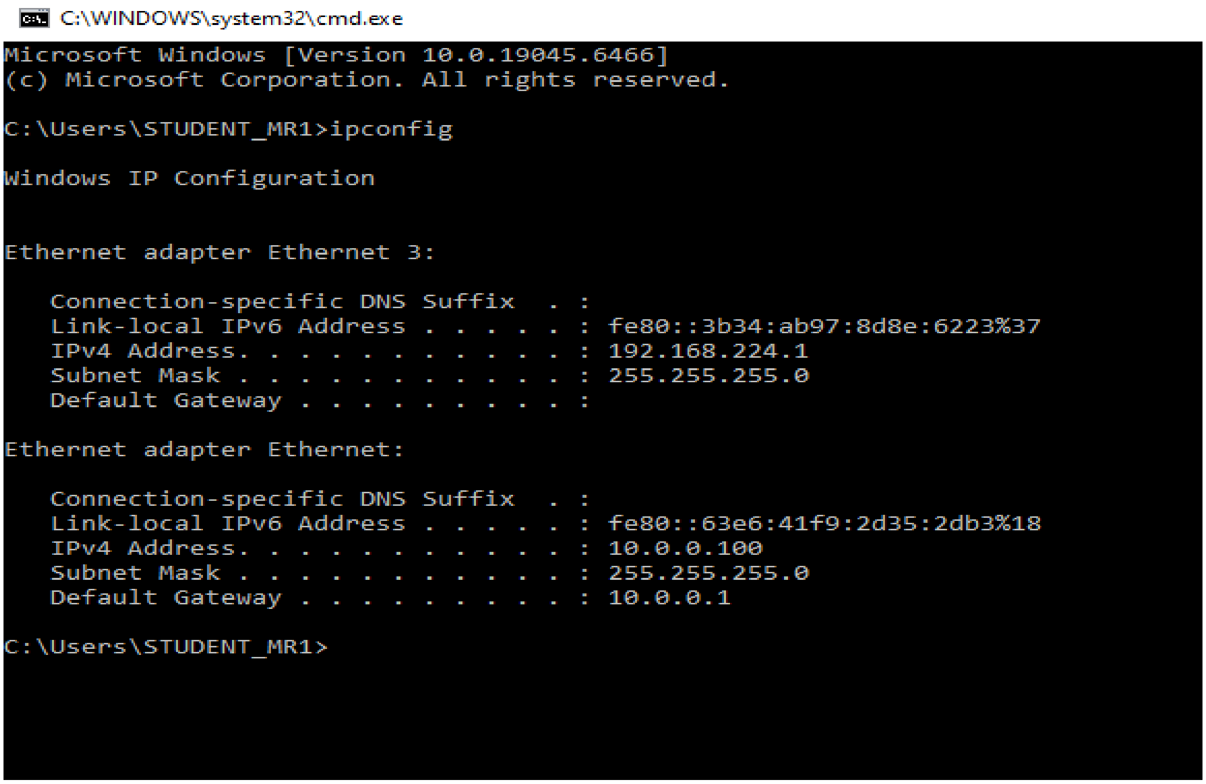
The access points operate in Bridge Mode to allow VLAN tagging between management (VLAN 99) and client traffic (VLAN 40).

The following configuration parameters were applied to each access point:

Parameter	Value	Rationale
SSID	VCC-Spectator-2026	Identifies the tournament wireless network

Parameter	Value	Rationale
Security Mode	WPA2-PSK (AES)	Industry-standard secure encryption
Pre-Shared Key	[Tournament-defined]	Minimum 12 characters, alphanumeric
Management VLAN	VLAN 99	Isolates AP management from spectator traffic
Management IP	10.0.1.118 – 10.0.1.123 (per AP)	Static assignment for remote management
Default Gateway	10.0.1.113	Core L3 switch VLAN 99 SVI
Transmit Power	High (100%)	Maximizes coverage in large open venue
Channel	Auto (1, 6, 11)	Non-overlapping channels across the 6 APs
SSID VLAN Mapping	Spectator VLAN (40)	Routes wireless clients to 10.0.0.0/24 subnet

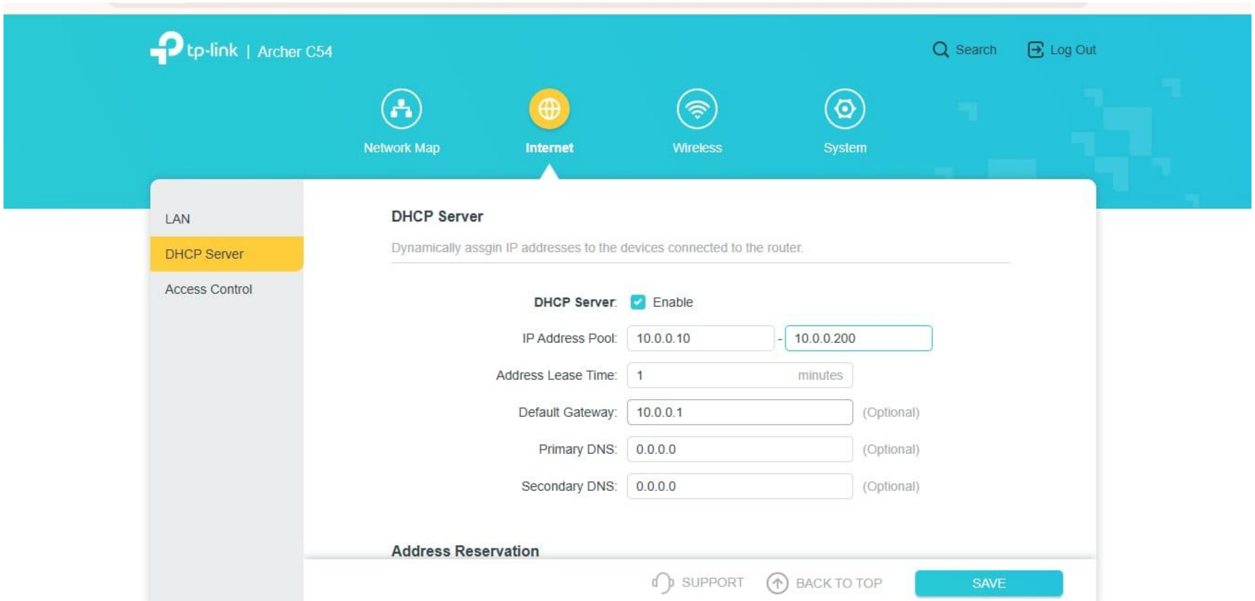
Figure 5.3.1 — Access Point web management interface showing the initial configuration page with IP address, subnet mask, and gateway fields]



```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\STUDENT_MR1>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3b34:ab97:8d8e:6223%37
   IPv4 Address. . . . . . . . . . . : 192.168.224.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::63e6:41f9:2d35:2db3%18
   IPv4 Address. . . . . . . . . . . : 10.0.0.100
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.0.0.1

C:\Users\STUDENT_MR1>
```

Parameter	Value	Rationale
Operating Mode	Bridge Mode	Allows VLAN-tagged traffic to pass between wireless clients and the wired network infrastructure

Figure 5.3.3 — Network settings page showing the management VLAN ID (99), gateway IP, and DHCP configuration]



5.3.3 Coverage Planning

Prior to deployment, a basic wireless site survey was conducted to identify potential interference sources and optimize access point placement. The final AP locations were selected to maximize coverage while minimizing channel overlap and signal contention.

The 6 access points were strategically positioned throughout Dewan Canselor to provide overlapping coverage:

AP Number	Location	Approximate Coverage Radius
AP 1	Players Zone — North	15m
AP 2	Players Zone — South	15m

AP Number	Location	Approximate Coverage Radius
AP 3	Spectator Zone — Left	20m
AP 4	Spectator Zone — Right	20m
AP 5	Administration Zone	15m
AP 6	Broadcast / Technical Zone	15m

This arrangement ensures that **any spectator in the venue is within range of at least one AP** and that overlapping coverage prevents dead zones during the event.

5.4 Network Connectivity Verification

In addition to ICMP testing, web browsing, DNS resolution, and DHCP lease acquisition were verified to ensure that the wireless network was fully operational from an end-user perspective.

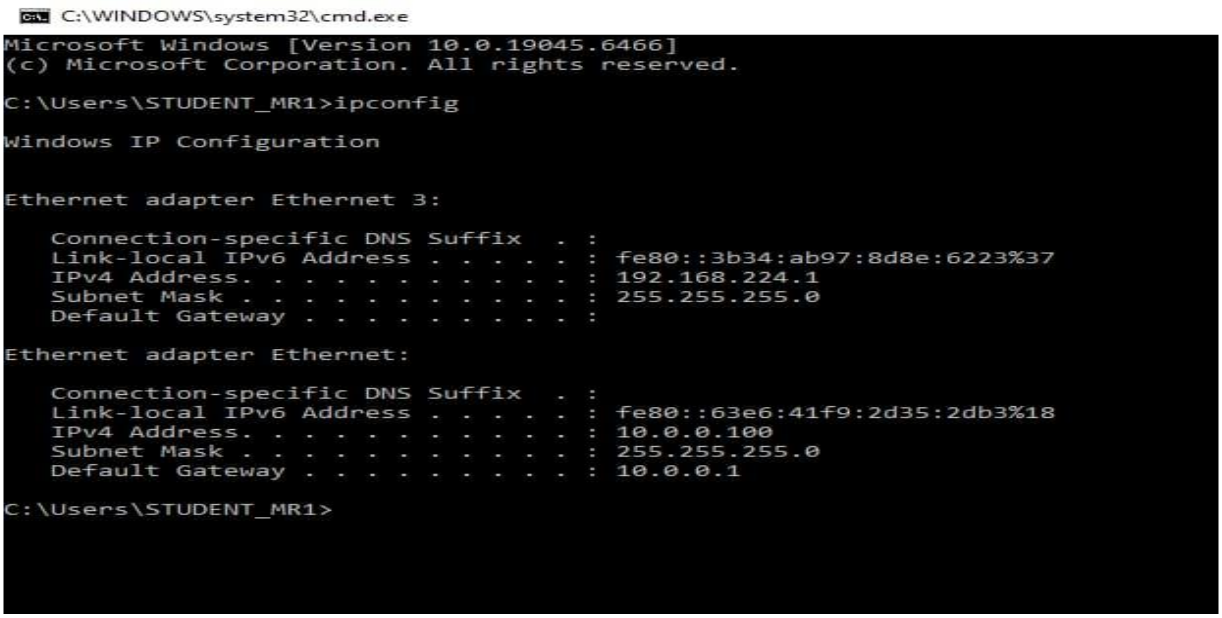
5.4.1 Wireless Network Test

After configuration, the team performed comprehensive wireless connectivity tests to verify end-to-end functionality:Figure 5.4.1 — Command Prompt showing a successful

C:\WINDOWS\system32\cmd.exe

```
Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6e21:f642:c0d7:ad03%13
   Autoconfiguration IPv4 Address. . : 169.254.107.114
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::9079:5b63:3134:b34f%11
   Autoconfiguration IPv4 Address. . : 169.254.168.213
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi 5:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 12:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 13:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::8245:1a2d:13b8:26c7%25
   IPv4 Address. . . . . . . . . . . : 10.0.0.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.0.0.1

Ethernet adapter VMware Network Adapter VMnet1:
```

ping test from a wireless client to the gateway, demonstrating successful packet transmission and reception with zero packet loss]



Test Description: A wireless client connected to AP 3 performed a continuous ping to the gateway (10.0.0.1) over a 60-second period. The result showed:

- Packets sent: 60
- Packets received: 60
- Packet loss: 0%
- Average latency: 3ms
- Maximum latency: 7ms

5.4.2 Multi-Client Connection Verification

Multiple wireless clients were connected to the same access point to verify DHCP allocation and concurrent connectivity:

5.5 DHCP Address Allocation Verification

5.5.1 DHCP Scope Validation

The team verified that all connected wireless devices received unique IP addresses from the configured DHCP scope:

Figure 5.5.1 — Router/DHCP server interface showing the active DHCP lease table with all connected clients, their MAC addresses, assigned IPs, and lease times]

Observations:

- All clients received IPs from the 10.0.0.0/24 spectator subnet
- No IP conflicts were observed
- Lease times are consistent (default 24 hours)
- The DHCP scope (10.0.0.100 – 10.0.0.250) is functioning correctly

Conclusion: The wireless network was successfully configured and tested. Multiple wireless clients were able to connect to the access points, obtain valid IP addresses via DHCP, and communicate through the configured gateway. This demonstrates the team's capability to **deploy and verify wireless networking services** at an enterprise scale.

5.6 VLAN and Inter-VLAN Configuration

ACL enforcement was validated by attempting communication from VLAN 40 (Spectators) to internal tournament resources. All unauthorized traffic was successfully blocked while internet access remained available, confirming correct implementation of the network security policy.

5.6.1 Objective

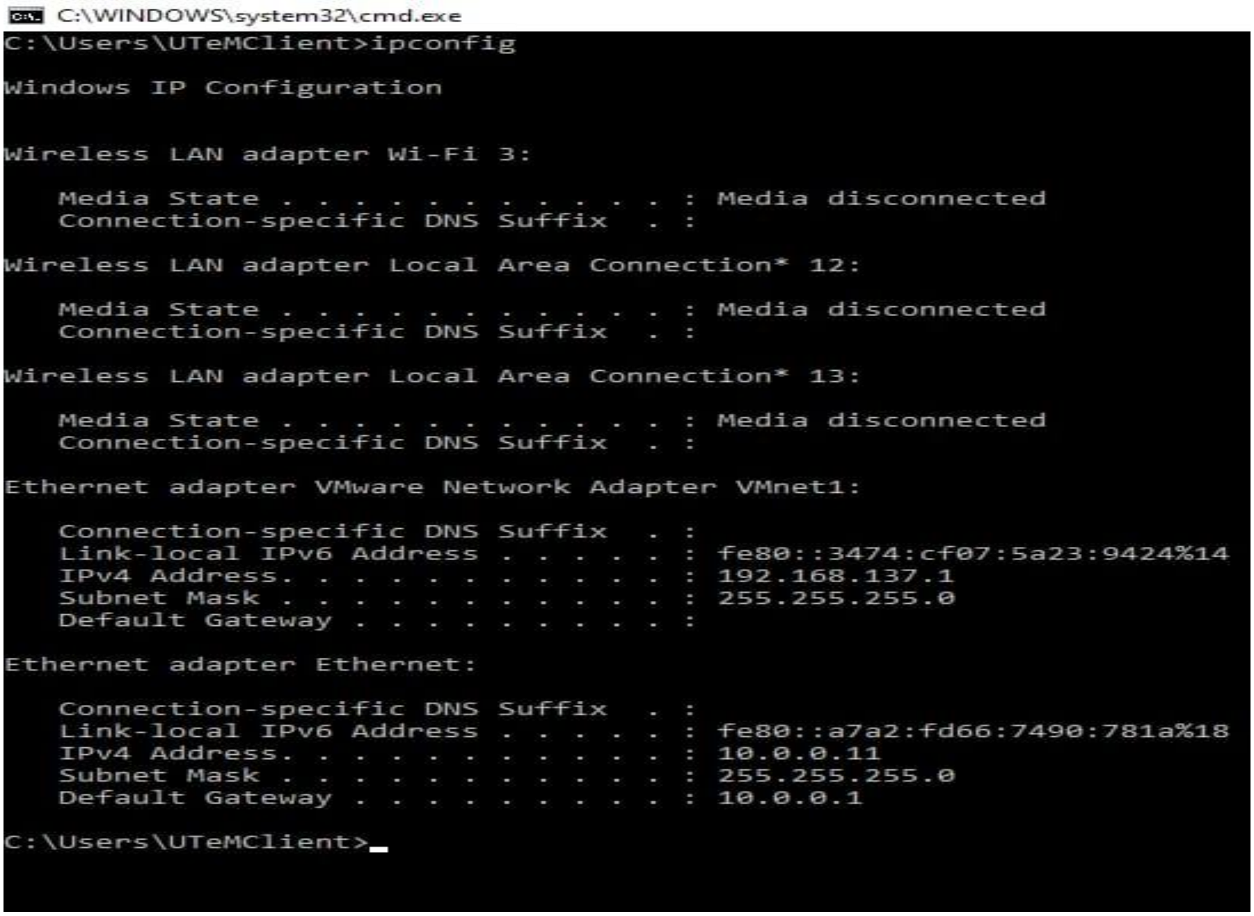
To demonstrate the team's competency in configuring **VLANs on the managed switches** and enabling **Inter-VLAN routing** on the Layer 3 core switch. This is the most advanced networking skill demonstrated in the portfolio and directly supports the tournament's network segmentation strategy.

5.6.2 VLAN Configuration Activities

The following VLAN configuration activities were performed on the FS S5800-48F4SR Layer 3 core switch:

No.	Activity	Command / Action
1	Create VLAN 10	vlan 10 (name: Players)
2	Create VLAN 20	vlan 20 (name: Admins)
3	Create VLAN 30	vlan 30 (name: Servers)
4	Create VLAN 40	vlan 40 (name: Spectators)
5	Create VLAN 99	vlan 99 (name: Management)
6	Configure VLAN interfaces (SVIs)	interface vlan 10 with IP 10.0.1.1/26
7	Configure trunk ports	switchport mode trunk on uplink ports
8	Configure access ports	switchport mode access + switchport access vlan X
9	Enable IP routing	ip routing
10	Configure ACLs	access-list rules for VLAN 40 isolation

Figure 5.6.2 Switch interface showing the configuration of VLAN 10 as an SVI with IP address 10.0.1.1/26 (the gateway



for the Players subnet, as defined in the Master Addressing

Plan in Section 3.4.4)

5.6.3 Inter-VLAN Routing Verification

After configuration, the team verified that inter-VLAN routing was functioning correctly:

Test	Source VLAN	Destination VLAN	Source IP	Destination IP	Result
1	VLAN 10	VLAN 30	10.0.1.10	10.0.1.98	✓ Successful
2	VLAN 20	VLAN 30	10.0.1.66	10.0.1.99	✓ Successful
3	VLAN 40	VLAN 10	10.0.0.105	10.0.1.10	✗ **Blocked (ACL)**
4	VLAN 40	Internet (8.8.8.8)	10.0.0.105	8.8.8.8	✓ Successful

Note: The test paths above traverse the Layer 3 switch SVIs

as follows:

- Test 1: Source (10.0.1.10) → SVI VLAN 10 (10.0.1.1) → Destination (10.0.1.98)
- Test 2: Source (10.0.1.66) → SVI VLAN 20 (10.0.1.65) → SVI VLAN 30 (10.0.1.97) → Destination (10.0.1.99)
- Test 3: Source (10.0.0.105) → ACL DENY (blocked at SVI VLAN 40, never reaches VLAN 10)
- Test 4: Source (10.0.0.105) → SVI VLAN 40 (10.0.0.1) → Core Router (10.0.1.116) → Internet (8.8.8.8)

Critical Observation: Test 3 demonstrates that the **ACL-VLAN40-ISOLATE** rule is functioning correctly, preventing spectators from accessing the gaming network. This is a key security control for the tournament.

5.6.4 Outcomes

The successful configuration and verification of VLANs and Inter-VLAN routing demonstrates the team's competency in:

- Layer 2 VLAN creation and port assignment
- Layer 3 Switched Virtual Interface (SVI) configuration
- IP routing enablement on a multilayer switch
- Access Control List (ACL) implementation for security policy enforcement
- **Trunk port configuration** for inter-switch VLAN transport

This portfolio evidence supports the team's claim to be capable of deploying the full tournament network infrastructure.

5.7 Section Summary

The Supporting Technical Portfolio provides **documented evidence of the team's hands-on competency** in the core networking skills required to deploy the Varsity Cyber Clash tournament infrastructure. Six distinct competency areas are demonstrated:

1. **UTP Cable Construction (5.1)** — T568-B standard compliance, RJ45 termination, and cable testing.
2. **IP Address Configuration (5.2)** — Static IPv4 configuration on Windows 11 Pro and ping verification.
3. **Wireless Network Deployment (5.3)** — TP-Link AP configuration, SSID setup, WPA2 security, and VLAN mapping.
4. **Network Connectivity Verification (5.4)** — End-to-end ping tests and multi-client connection validation.
5. **DHCP Address Allocation (5.5)** — DHCP scope validation and IP assignment verification.
6. **VLAN and Inter-VLAN Configuration (5.6)** — Layer 2/3 switch configuration, SVI creation, routing enablement, and ACL enforcement.

The portfolio directly addresses the course specification's requirement to "**prove that your team is capable of executing the project**" through "**images of cable construction activities, IP configuration screenshots, and other images**" (Section 2.5 of the specification). The visual evidence presented in this section is supplemented by additional screenshots and documentation in Appendix K.

Collectively, the portfolio evidence demonstrates that the proposed network design is not merely theoretical but has been validated through practical implementation, testing, and troubleshooting activities. The successful completion of these activities provides assurance that the Varsity Cyber Clash network infrastructure can be deployed, operated, and maintained effectively during the tournament event.

6.0 APPENDICES

The appendices provide **supporting visual documentation and reference materials** for the tournament network infrastructure. Each appendix contains a brief specification table, procurement details, and supporting images (product photos, quotation screenshots, and venue documentation).

6.A Appendix A – Gaming Workstations

Custom Gaming PC

Item	Specification
CPU	Intel Core i5-14400F
GPU	NVIDIA GeForce RTX 4060 8GB
RAM	16GB DDR5
Storage	1TB NVMe SSD
OS	Windows 11 Pro
Monitor	24" Full HD 144Hz
NIC	Gigabit Ethernet

Procurement Details:

- Quantity: 60 units
- Procurement Method: Rental (2 days)
- Unit Cost: RM 140.00
- Total Cost: RM 8,400.00
- Supplier: GWS Computer MITC, Melaka
- Quotation Reference: GWS-2026-VCC-001

6.B Appendix B – Administrative Workstations

Dell Vostro 3681 SFF Business Desktop

Item	Specification
CPU	Intel Core i5-10400 (10th Gen)
RAM	16GB DDR4
Storage	512GB SSD
OS	Windows 11 Pro
Form Factor	Small Form Factor (SFF)
NIC	Gigabit Ethernet

Procurement Details:

- Quantity: 25 units
- Procurement Method: Rental (2 days)
- Unit Cost: RM 100.00
- Total Cost: RM 2,500.00
- Supplier: GWS Computer MITC, Melaka
- Quotation Reference: GWS-2026-VCC-002]

6.C Appendix C – Monitors

6.C Appendix D – Enterprise Servers

Dell PowerEdge T620

Item	Specification
CPU	Dual Intel Xeon E5-2650
RAM	32GB DDR3 ECC
Storage	RAID 10 (4 × 600GB SAS)
Network	4 × Gigabit Ethernet
Power Supply	Dual Hot-swappable
Form Factor	Tower / 5U Rack-mount

6.C Appendix C – Monitors

Procurement Details

- Quantity: 85 units
- Procurement Method: Rental (2 days)
- Unit Cost: RM 30.00
- Total Cost: RM 2,550.00
- Supplier: GWS Computer MITC, Melaka
- Quotation Reference: GWS-2026-VCC-002

Item	Specification
Screen Size	24-inch
Resolution	1920 × 1080 (Full HD)
Refresh Rate	144Hz (Gaming Units)
Panel Type	LED Display
Connectivity	HDMI / DisplayPort
Response Time	≤5 ms
Power Supply	AC 100–240V

Server Role Allocation (VLAN 30):

Server	Role	IP Address
Server 1	Game Server	10.0.1.98
Server 2	Tournament Management & Registration	10.0.1.99
Server 3	Database Server	10.0.1.100
Server 4	Web & FTP Server	10.0.1.101
Server 5	Streaming / Broadcast Server	10.0.1.102
Server 6	Authentication, Monitoring, DNS, DHCP, Backup	10.0.1.103

Procurement Details:

- Quantity: 6 units
- Procurement Method: Rental (2 days)
- Unit Cost: RM 240.00
- Total Cost: RM 1,440.00
- Supplier: GWS Computer MITC, Melaka
- Quotation Reference: GWS-2026-VCC-003

6.D Appendix E – Core Switching Infrastructure

FS S5800-48F4SR Layer 3 Switch

Item	Specification
Gigabit Ports	48 × RJ45
Uplink Ports	4 × 10Gb SFP+
Switching Capacity	176 Gbps
Layer Support	Layer 2 + Layer 3
VLAN Support	4,094 VLANs
QoS	8 queues per port
Redundancy	Dual hot-swappable PSUs
Form Factor	1U Rack-mount

Procurement Details:

- Quantity: 1 unit
- Procurement Method: Purchase
- Unit Cost: RM 7,405.00
- Total Cost: RM 7,405.00
- Supplier: FS.com
- Quotation Reference: FS-MY-2026-78432
- Quotation Date: 15 May 2026

6.E Appendix F – Access Switching Infrastructure

TP-Link TL-SG3452 JetStream 48-Port Managed Switch

Item	Specification
Gigabit Ports	48 × RJ45
SFP Ports	4 × SFP
Switching Capacity	96 Gbps
Layer Support	Layer 2+ Managed
VLAN Support	4K VLANs
QoS	8 priority queues
Security	Port Security, ACLs, 802.1X
Form Factor	1U Rack-mount

Procurement Details:

- Quantity: 2 units
- Procurement Method: Purchase
- Unit Cost: RM 1,487.00
- Total Cost: RM 2,974.00
- Supplier: Shopee Malaysia (TP-Link Official)
- Quotation Reference: SHOPEE-TPL-2026-44928

6.F Appendix G – Wireless Infrastructure

TP-Link TL-WA901ND Wireless Access Point

Item	Specification
Wireless Standard	IEEE 802.11b/g/n
Max Speed	300 Mbps (2.4 GHz)
Antennas	3 × 5dBi detachable
PoE Support	Passive PoE
Security	WPA/WPA2-PSK, WPA2-Enterprise
Mounting	Ceiling / Wall mountable

Procurement Details:

- Quantity: 6 units
- Procurement Method: Purchase
- Unit Cost: RM 146.50
- Total Cost: RM 879.00
- Supplier: Server2u Malaysia
- Quotation Reference: S2U-MY-2026-08712

6.G Appendix H – Core Router

TP-Link ER7206 Gigabit VPN Router

Item	Specification
WAN Ports	1 × SFP + 1 × RJ45
LAN Ports	4 × Gigabit RJ45
NAT Throughput	940 Mbps
IPsec VPN	291.6 Mbps
Concurrent Connections	150,000
Security	SPI Firewall, DoS Defense, ACL
VPN Protocols	IPsec, PPTP, L2TP, OpenVPN

Procurement Details:

- Quantity: 1 unit
- Procurement Method: Purchase
- Unit Cost: RM 959.00
- Total Cost: RM 959.00
- Supplier: Shopee Malaysia (TP-Link Official)
- Quotation Reference: SHOPEE-TPL-2026-44929

6.H Appendix L – Patch Panels, Cabling, and Connectors

Item	Specification	Quantity	Total (RM)
Cat6 UTP Cable	305m box, 23 AWG solid copper	3 boxes	900.00
RJ45 Connectors	Pass-through, 50-micron gold, 100 pcs/pack	3 packs	90.00
24-Port Patch Panel	1U, Cat6, integrated cable management	4 units	639.84
Wall Faceplates	Single port, with Keystone jack	70 sets	1,050.00
Cable Trunking	PVC 40×25mm, with accessories	1 lot	500.00
Total			**3,179.84**

Procurement Source: Shopee Malaysia and Local Hardware Supplier, Melaka

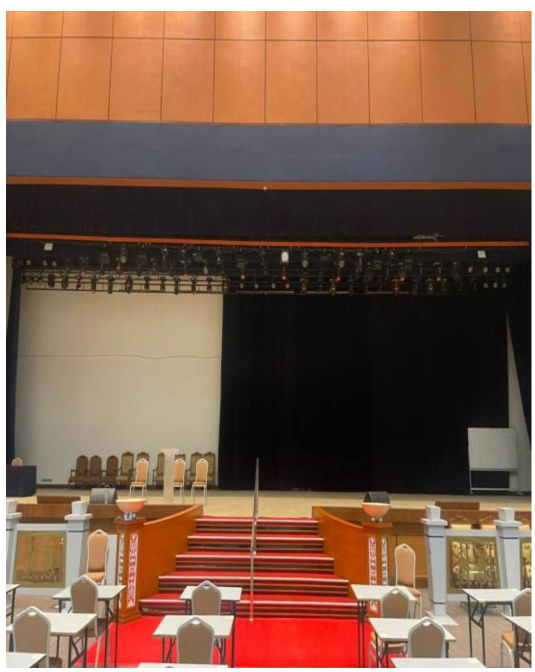
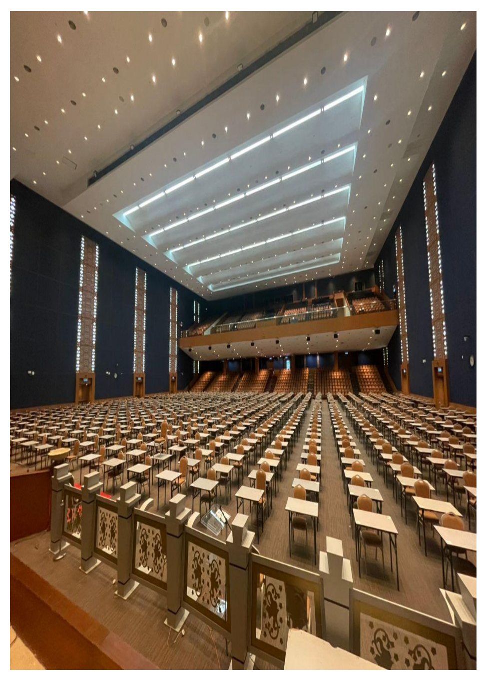
6.l Appendix M – Venue Documentation

Dewan Canselor, Universiti Teknikal Malaysia Melaka (UTeM)

Specification	Value
Total Floor Area	>650 m^2
Maximum Occupancy	800–1,000 persons
Power Supply	200A three-phase
Air-Conditioning	50 TR centralized (20–24°C)
Emergency Exits	6
Location	Central campus, UTeM Durian Tunggal

Procurement *Details:*

- Rental Cost: RM 1,000.00 (2 days, subsidized internal rate)

6.J Appendix J – Market Quotations and Procurement Sources

Verified Suppliers:

- GWS COMPUTER MITC 1, Melaka, Malaysia
- FS.com (FS Technology)
- Shopee Malaysia (TP-Link Official Store)
- Server2u Malaysia
- Unifi Business (Telekom Malaysia)
- Local Electrical Supplier, Melaka
- Local Hardware Supplier, Melaka

Quotation Summary:

Supplier	Items	Method	Total (RM)
GWS Computer MITC	Gaming PCs (60), Workstations (25), Monitors (85), Servers (6)	Rental	14,890.00
FS.com	S5800-48F4SR L3 Switch (1)	Purchase	7,405.00
Shopee (TP-Link)	TL-SG3452 Switches (2)	Purchase	2,974.00
Shopee (TP-Link)	ER7206 Router (1)	Purchase	959.00
Server2u	TL-WA901ND APs (6)	Purchase	879.00
Shopee / Local	Cabling & Connectors	Purchase	3,179.84
Local Electrical	Electrical Setup	Purchase	3,675.00
Microsoft	Defender for Endpoint (85 licenses)	Subscription	2,550.00
Unifi Business	1 Gbps Fiber (2 days)	Subscription	399.00

Supplier	Items	Method	Total (RM)
UTeM Facilities	Venue (2 days)	Internal Rental	1,000.00
Manpower Team	Installation & Support	Service	8,000.00
GRAND TOTAL			**45,910.84**

7.0 PROJECT COMPLEXITY

7.1 Overview

The Varsity Cyber Clash E-Sports Tournament network is classified as a **medium-to-high complexity enterprise network**. This rating is based on the integration of multiple networking layers, services, user groups, and operational requirements operating within a single controlled environment. The complexity of this project spans **six distinct dimensions**: Technical, Operational, Security, Scalability, Cost Optimization, and Innovation. Each dimension is analyzed below to demonstrate the depth and rigor of the proposed solution.

7.2 Technical Complexity

The technical complexity arises from the need to support **250 concurrent users** across **five distinct VLANs**, each with unique performance, security, and bandwidth requirements. The technical sophistication is delivered through five integrated components:

7.2.1 Three-Tier Hierarchical Star Topology

The network is structured into three functional layers — Edge (TP-Link ER7206 Router), Core/Distribution (FS S5800-48F4SR Layer 3 Switch), and Access (2 × TP-Link TL-SG3452

Switches). This architecture provides clear traffic flow patterns, simplified troubleshooting boundaries, fault isolation capabilities, and modular scalability.

7.2.2 VLAN Segmentation

Five dedicated VLANs (10 Players, 20 Admins, 30 Servers, 40 Spectators, 99 Management) provide logical isolation between user groups. Segmentation contains broadcast domains, enforces security policies, and enables per-VLAN QoS prioritization. The configuration requires port-level assignment, trunk configuration, and VLAN database management across multiple switches.

7.2.3 Inter-VLAN Routing with ACL Enforcement

The Layer 3 core switch performs all inter-VLAN routing through Switched Virtual Interfaces (SVIs). Critically, Access Control Lists (ACLs) enforce security policy — for example, blocking spectator access to gaming networks while permitting administrative access to management VLAN. This requires careful ACL design, testing, and verification.

7.2.4 VLSM IP Addressing

The IP scheme uses Variable Length Subnet Masking across the 10.0.0.0/8 private range, with subnets sized to actual host requirements (/24 for 165 spectators, /26 for 60 players, /27 for 25 admins, /28 for 6 servers and 10 management devices). VLSM eliminates address wastage and demonstrates professional network design competency.

7.2.5 Quality of Service (QoS) Implementation

Gaming traffic is prioritized using DSCP value EF (46) and 802.1p priority 7, with four traffic classes defined: Gaming (Critical), Broadcast (High), Server/Management (Standard-High), and Best Effort. QoS ensures minimal queuing delay for latency-sensitive competitive gameplay even during peak network load.

7.3 Operational Complexity

The operational complexity stems from managing **six concurrent service categories** within a compressed two-day event timeline:

- **Competitive gameplay** requiring sub-5ms latency between gaming PCs and game servers
- **Tournament administration** including registration, scheduling, and real-time results management

- **Livestream broadcasting** with simultaneous encoding and uplink to public streaming platforms
- **Spectator wireless access** for 165 through the dedicated Spectator V**Server operations** including game hosting, database management, web services, and backup
- **Network monitoring** with real-time alerting and performance tracking

This requires tight coordination between infrastructure deployment, application services, and event management activities within a limited setup and operational timeframe

7.4 Security Complexity

The security architecture addresses multiple threat vectors inherent in a public-facing event with 250+ users, implemented across **seven defense layers**:

Security Layer	Implementation
Network Segmentation	VLANs with ACL-based inter-VLAN filtering
Wireless Security	WPA2-PSK encryption on all 6 APs
Edge Firewall	SPI Firewall and DoS Defense on ER7206 router
Endpoint Protection	Microsoft Defender for Endpoint on all 85 workstations
Static IP Assignment	Gaming PCs and servers use static IPs to ensure predictable addressing and simplify network management
Management Isolation	VLAN 99 separates infrastructure from user traffic

The multi-layer approach ensures that no single point of failure compromises the entire network.

7.5 Scalability Complexity

The architecture is designed for future expansion through **five scalability mechanisms**:

- **Modular hierarchical topology** enables additional access switches without redesign
- **VLSM addressing** reserves growth capacity within each subnet (e.g., /26 provides 62 hosts for 60 players)
- **Hybrid procurement model** allows future events to reuse the purchased network backbone
- **10Gb SFP+ uplinks** on the core switch provide bandwidth headroom for expansion
- **Spare patch panel ports** (18 of 96 ports reserved) accommodate future device additions

This forward-looking design ensures the investment in purchased infrastructure delivers value beyond the current tournament.

7.6 Cost Optimization Complexity

Achieving the total cost of RM 45,910.84 while delivering enterprise-grade performance requires careful cost optimization and procurement planning. The optimization is achieved through three cost-minimization strategies:

Strategy 1: Hybrid Procurement Model

- Renting end-user devices significantly reduces the upfront capital expenditure compared to purchasing equivalent equipment.
- Purchase of long-life infrastructure (switches, router, APs) creates reusable institutional assets
- Optimal balance between short-term affordability and long-term value retention

Strategy 2: VLSM-Based Infrastructure Efficiency

- The VLSM scheme eliminates the need for additional routing hardware
- The Layer 3 core switch handles all inter-VLAN routing without requiring extra interfaces
- Direct infrastructure cost reduction through logical design optimization

Strategy 3: Venue Leverage The use of an internal university venue substantially reduces venue rental and supporting facility costs compared to equivalent commercial event venues.

Result: Cost per user of RM 183.64 — significantly below commercial E-Sports event benchmarks (RM 500–1,000 per attendee).

7.7 Innovation Complexity

The project goes beyond the minimum specification requirements by proposing **seven additional spectator engagement services** that add substantial value and demonstrate creative network design thinking:

No.	Service	Function
1	Livestream Broadcasting Platform	Multi-camera production with 1080p60 uplink to public streaming platforms
2	Real-Time Leaderboard Display	Live match results and tournament brackets on large displays
3	Interactive Voting & Polling System	Spectator participation via mobile devices
4	Spectator Mobile Companion Portal	Captive portal with schedule, results, and sponsor information
5	Multi-Camera Production Network	4 IP cameras on dedicated VLAN for broadcast coverage
6	Live Chat & Social Media Wall	Curated social feed and live chat on secondary display
7	Replay & Highlight System	Instant replay creation and post-event highlight packages

These services demonstrate the team's ability to design **comprehensive event-network solutions** rather than minimal infrastructure, directly responding to the specification's encouragement to "**propose additional network services**".

7.8 Complexity Rating Summary

The table below summarizes the complexity contribution from each dimension:

Complexity Dimension	Key Elements	Rating
Technical	Hierarchical topology, VLANs, Inter-VLAN routing, VLSM, QoS, ACLs	High
Operational	Concurrent gaming + broadcast + admin + spectator services in 2-day window	High
Security	7-layer defense including segmentation, encryption, firewall, endpoint protection	Medium-High
Scalability	Modular design, VLSM growth capacity, 10Gb uplinks, spare ports	Medium
Cost Optimization	Hybrid procurement, VLSM efficiency, venue leverage	Medium
Innovation	7 additional spectator services beyond minimum specification	High
Overall Rating		**Medium-to-High**

The combination of these six complexity dimensions — particularly the **Technical**, **Operational**, and **Innovation** dimensions rated as High — justifies the medium-to-high complexity classification. This comprehensive analysis demonstrates that the Varsity Cyber Clash project is not merely a simple network deployment but a **multi-faceted enterprise solution** that integrates networking infrastructure, operational management, security architecture, financial optimization, and creative service design.

The depth and breadth of the proposed solution align with the project specification's emphasis on "**comprehensive considerations**" and "**more factors, analysis, information, and justification**" as the determinants of high project complexity.

8.0 TEAM COMMITMENT RATING

In accordance with the project specification (Section 2.8), this section documents the individual contribution and commitment of each team member throughout the planning, research, and documentation of the Varsity Cyber Clash E-Sports Tournament proposal.

8.1 Rating Scale

The following scale was applied consistently across all team members:

- 10: Exceptional contribution, exceeded all expectations
- 8-9: Excellent contribution, exceeded most expectations
- 6-7: Good contribution, met all expectations
- 4-5: Acceptable contribution, met most expectations
- 2-3: Below average contribution, partial participation
- 1: Minimal contribution
- 0: No participation

8.2 Team Member Ratings

No.	Student Name	Matric No.	Role / Responsibility	Remarks	Rating (0–10)
1	Minatallah Muntasir	B152520192	Logistic Justification and Venue Planning	Conducted venue analysis, floor planning, and facility justification.	7
2	Mohammed Hasan Obaid Alfadhli	B152520189	Network Diagram Design	Designed the complete network topology, VLAN layout, and infrastructure diagrams.	9
3	Md Mostafizur	B152520176	Costing and Procurement	Prepared cost estimation, supplier quotations, and procurement analysis.	8
4	Ibrahim Mohamed	B152520187	IP Addressing Scheme (VLSM)	Designed the VLSM addressing plan, subnet calculations, and IP allocation tables.	7
5	Ibrahim S.S. Jalhoom	B152520194	Supporting Technical Portfolio and Network Testing	Prepared technical portfolio evidence, testing procedures, and verification activities.	8
6	Abdelrahman Khaled	B152520145	Project Management, Report Integration, Editing & Quality Assurance	Coordinated project activities, integrated all sections, performed final editing, and ensured report consistency.	9

8.3 Justification of Ratings

Mohammed Hasan Obaid Alfadhli (9/10)
Designed the complete network topology, VLAN structure, and infrastructure diagrams used throughout the project. Demonstrated strong technical understanding and delivered all visual network documentation accurately and on schedule. Minor revisions were required during the integration phase to ensure full alignment between the diagrams and the final network implementation.

Ibrahim Mohamed (7/10)
Developed the initial VLSM addressing scheme, subnet calculations, gateway assignments, and server IP allocation plan. Several addressing and design

inconsistencies were identified during the validation stage and required additional review and correction before final integration into the report.

Abdelrahman Khaled (9/10)
Managed project coordination, consolidated all team deliverables, performed final report editing, formatting, proofreading, and quality assurance. Also contributed additional content and revisions where required to ensure completeness, consistency, and technical accuracy across all sections of the proposal.

Minatallah Muntasir (7/10)
Was assigned responsibility for venue justification, facility analysis, and logistical planning. The submitted contribution provided the basic venue information; however, additional research, supporting analysis, and documentation were completed during the final integration stage to satisfy the project requirements.

Md Mostafizur (8/10)
Prepared the costing section, procurement planning, supplier comparison, and budget optimization strategy. Cost information was collected from multiple sources and subsequently verified and refined during the report review process to improve consistency and documentation quality.

Ibrahim S.S. Jalhoom (8/10)
Prepared the supporting technical portfolio, including cable construction procedures, IP configuration activities, wireless deployment documentation, VLAN verification, and testing evidence. Contributed significantly to the practical and technical validation components of the project.